

Tournament score sequences and weighted component asymptotics

Explicit corrections and uniform inverse models

Research report 2 October 2026

Part II: conditional phase coexistence 3 October 2026

Abstract

The exact generating function for tournament score sequences permits a clean separation between an explicit square-root singularity and an analytic divisor remainder. We use that separation to derive computable asymptotic corrections for all and strong score sequences, and a uniform expansion for the generating function weighted by the number of irreducible components. The latter is valid to every fixed half-power order on a specified logarithmic subcritical window, with an absolute remainder. Its first two terms yield an explicit coexistence profile with a $\log \log n$ centering correction. We also give inverse-location results for specified one-variable models and, when n is known, for component-weight calibration. Exact small-size enumeration, symbolic coefficient identities, numerical comparisons, and portable replay code accompany the proof.

The leading enumeration theorem, exact generating function, and general renewal and heavy-traffic mechanisms are established prior work. The emphasis here is the explicit local lattice specialization, correction generators, uniform error bounds, and carefully delimited inverses. No exhaustive novelty claim is made.

Part II (added 3 October 2026) answers the first further question of Part I. Under a component weight approaching criticality at the coexistence rate, the score sequence is asymptotically a logistic mixture of a giant-block regime and a regime with a linear number of blocks. On the giant event the remaining blocks are, up to total variation $O(1/\log n)$, two independent geometric renewal strings, which gives $\text{Gamma}(2, 1)$ laws for the block count and the giant's deficit, stable residual fluctuations, a mixed Fréchet law for the second-largest block and every fixed inverse-logarithmic correction; on the other event the centered block count has a $3/2$ -stable limit jointly with a Fréchet largest block. Its coefficient input re-proves Part I's uniform two-term theorem; the probabilistic results are new to the repository.

Results and scope at a glance

- All scores are OEIS A000571; strong scores are A351822. Strongly connected *labeled tournaments*, A054946, are a different counting problem.
- At each fixed order K , all and strong counts have explicit relative corrections through n^{-K} with remainder $O(n^{-K-1})$.
- For $0 \leq \tau \leq C \log n$ with fixed C , the normalized weighted count has computable terms $n^{-m/2} C_m(\tau)$ and an absolute $O_{M,C}(n^{-(M+1)/2})$ remainder at every fixed M .
- Two coefficient contributions coexist at $\tau = \frac{1}{2} \log n + 2 \log \log n + \log(\mu\sqrt{\pi}/2) + s$. This is a coefficient asymptotic, not a proved conditional distributional mixture.
- The inverses specify the model, normalization, and known quantities. They do not certify finite- n integer rounding or finite- n confidence intervals.

The results do not cover $\tau < 0$, a supercritical moving pole, truncation order growing with n , or convergence of the full asymptotic series. All numerical decimal evaluations are non-interval computations.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 68 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `tournament-score-sequences-source.zip` (wrapper directory `tournament-score-sequences/`) in commit `096ee7b87`, and placed in commit `34f1acd4b` in the research-report collection, among the OEIS-sequence asymptotics reports (main file `report.tex`, now `article.tex`). The manuscript names no ProveIt commit and continues no repository path, so it has no pin; it names neither an author nor a tool, and its PDF metadata leave the author empty. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No other report of the repository treats tournament score sequences or A000571, A351822, A145855, A054946. The writing step prefixed every label with `tss:`, typeset the six binomial coefficients with `\binom` instead of `\choose` (same output, no amsmath warning), set the bibliography ragged-right, and added these [write] notes; no statement, proof or number of the manuscript was altered.

Credit, kept. The exact generating function (3) and the block decomposition are Claesson–Dukes–Franklín–Stefánsson’s [1] (with the divisor evaluation attributed there to Alekseyev), the leading asymptotics (8) are Kolesnik’s [2] (shorter proof [3]), and the renewal and logarithmic heavy-traffic/heavy-tail mechanisms are Blanchet–Glynn’s and Olvera-Cravioto–Blanchet–Glynn’s [4, 5]. The report’s contribution is the explicit local refinement; no exhaustive novelty search is claimed.

Notation. The manuscript reuses several letters. Each symbol is printed as delivered; the table fixes its meaning by section, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
W, W_{-1}	$W(z, u)$ is the component-weighted generating function and $W_n(u)$ its coefficients; W_{-1} in (51) is the lower branch of Lambert’s function.
s	$s = nw$ is the Hankel variable (Sections 4–5); s is also the shift of the coexistence centre, $\tau = b_n + s$ (Sections 6–7).
b_j, b_n, B	b_j are the odd Puiseux coefficients of $B(t)$ (15); b_n is the coexistence centre (47); $B_j(x)$ are Bernoulli polynomials.
c, c_m^σ, C	$c = e^{-\lambda}\mu$; c_m^σ the fixed-order corrections; c also a contour constant in $\exp(-cn w)$. C is the fixed window constant ($0 \leq \tau \leq C \log n$); $C_m(\tau)$ the uniform terms; $C_\sigma = e^{\sigma\lambda}/(2\sqrt{\pi})$.
D_j^σ, D, d_j, d	D_j^σ the odd local coefficients (21); $D(w)$ the denominator series (40) with coefficients d_j ; d a divisor in (2). $d_4 = \ell$.
E, e_j, E_n	$E(v) = \sum e_j v^j$ is the even part (14); $E_n(w)$ the resolvent error (35). e_2 is a constant, not e^2 .
$H, H_j, \mathcal{H}, h$	$H(\tau)$ the entire kernel (32); H_j harmonic numbers in (16); $\mathcal{H}$ the Hankel contour; $h = \log 4$ (Section 7).
k, K	$k = 2/(3\mu)$; k is also an index. K is a fixed truncation order (Section 3) and a generic constant; K_s the path constant (54).
p, q, q_j	$p = 1 - e^{-\lambda}$; $q = \tau/n$; q_j the renewal mass function.
t	$t = \sqrt{1 - 4z}$ (Sections 2–3); t is the free weight parameter of $\mathcal{M}_{n,M}(t)$ in Section 7.
F, R, r_n	$F(z)$ the central-binomial part and $R(z)$ the divisor remainder (10), $r_n = [z^n]R$; $F_{K,\sigma}(x)$ the fixed-order inverse model.
a, a_c, a_n	$a = a_n(\tau) = u_n(\tau)c$; $a_c = c/p$ its critical value; $a_x(\tau_x)$ its real-variable extension in (52).
m, M	m counts strong blocks in $S_{n,m}$ and indexes c_m^σ, C_m ; M is a truncation order and also a summation cutoff in (20).
I, S, N_n	$I(z)$, I_n strong sequences; $S(z)$, S_n all sequences, $S_{n,m}$ by number of blocks; N_n the divisor sum (A145855). I_n omits A351822’s empty object.

[write] **Added 3 October 2026, batch 85.** Since 3 October 2026 this report has two Parts. Part I is the batch-77 manuscript described in the preceding note, printed as before: its sections, equations, theorems and labels keep their numbers, so “Section 6” and “Section 9” still mean what the manuscript of Part II cites. The preceding note’s “single manuscript, printed in full” now describes Part I; its statement that no other report treats tournament score sequences remains true. Part II (Sections 10–20 and two appendices) is manuscript 04

of batch 85, an AI-assisted, unrefereed and unformalized manuscript pinned at commit `d68b65ea0`. It answers Part I's first further question: at the weight $u_n(b_n + s)$ the structure is a logistic mixture, with probability tending to $(1 + e^{-s})^{-1}$ of one block of size $n - O_{\mathbb{P}}(n/\log n)$ and otherwise $n/a_c + O_{\mathbb{P}}(n^{2/3})$ blocks and no macroscopic block, and it derives conditional Gamma, stable, Fréchet and point-process laws, residual approximations to every fixed inverse-logarithmic order, and an $O(1/\log n)$ rate (Theorems 12.1–12.4, Proposition 18.1). The scope bullet above that calls the coexistence profile “a coefficient asymptotic, not a proved conditional distributional mixture” remains true of Corollary 6.1 by itself; the mixture is Part II's theorem. Part II's symbols m , a_n , r_n , k , K , d and B do not mean what they mean here; a separate table at the start of Part II fixes them.

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Part I

Explicit corrections and uniform inverse models

1 The counting problem and established identities

A tournament is an orientation of a complete graph. Its score sequence is the nondecreasing list of vertex outdegrees. Let S_n count distinct score sequences of length n , and set $S_0 = 1$. A sequence $(s_1, \dots, s_n)$ is a score sequence precisely when

$$0 \leq s_1 \leq \dots \leq s_n \leq n-1, \quad \sum_{i=1}^j s_i \geq \binom{j}{2} \ (j < n), \quad \sum_{i=1}^n s_i = \binom{n}{2}. \quad (1)$$

The sequence is strong or irreducible when each proper-prefix inequality is strict. This includes the singleton sequence. We use I_n for the number of nonempty strong sequences, with $I_0 = 0$. The OEIS entry A351822 includes an empty-object value 1 at index zero; its full listed generating function is consequently $1 + I(z)$ under our notation.

Equality indices in (1) give the unique ordered decomposition into strong blocks. When a prefix of length j is removed, subtract j from each subsequent score. Conversely, concatenating two blocks after adding the first length to the second block gives a score sequence. Thus a block is an object of the score-sequence class, not an underlying labeled tournament. The first values are

n	0	1	2	3	4	5	6	7	8	9	10
S_n	1	1	1	2	4	9	22	59	167	490	1486
I_n	0	1	0	1	1	3	7	21	61	184	573

Let φ be Euler's totient function and define

$$N_n = \frac{1}{2n} \sum_{d|n} (-1)^{n+d} \varphi(n/d) \binom{2d}{d}, \quad A(z) = \sum_{n \geq 1} \frac{N_n}{n} z^n. \quad (2)$$

The auxiliary N_n is OEIS A145855. Claesson, Dukes, Franklin, and Stefánsson (CDFS) prove the score-sequence identity [1, Corollaries 12–13]

$$S(z) = \sum_{n \geq 0} S_n z^n = \exp A(z). \quad (3)$$

Their paper attributes the divisor evaluation in (2) to Alekseyev's 2008 proof of a conjecture of Jovović. The signs $(-1)^{n-d}$ and $(-1)^{n+d}$ are equal. These established exact identities are inputs to the analytical results below; we do not reproduce the full combinatorial proof of (3).

The ordered block decomposition now gives

$$I(z) = 1 - \exp(-A(z)), \quad W(z, u) = \frac{1}{1 - uI(z)} = \sum_{n, m \geq 0} S_{n,m} z^n u^m. \quad (4)$$

Here $S_{n,m}$ counts sequences with m strong blocks, $S_{0,0} = 1$, and $W_n(u) = [z^n]W(z, u)$. For integer $u > 0$, the factor u^m colors the blocks; for real $u > 0$ it is a positive component weight. In particular $W_n(1) = S_n$ and $S_{n,1} = I_n$.

Exact computation uses

$$nS_n = \sum_{k=1}^n N_k S_{n-k}, \quad (5)$$

$$I_n = S_n - \sum_{k=1}^{n-1} I_k S_{n-k}, \quad (6)$$

$$W_0(u) = 1, \quad W_n(u) = u \sum_{k=1}^n I_k W_{n-k}(u). \quad (7)$$

The first two recurrences can be evaluated with integers. Independently enumerating the solutions of (1) checks both the totals and the number of equality indices, hence the component statistic itself.

1.1 Prior asymptotics and the intended refinement

Kolesnik [2] proved the leading asymptotics for S_n , I_n , and each fixed component count. In the notation introduced below, these include

$$S_n \sim \frac{e^\lambda}{2\sqrt{\pi}} \frac{4^n}{n^{5/2}}, \quad I_n \sim \frac{e^{-\lambda}}{2\sqrt{\pi}} \frac{4^n}{n^{5/2}}, \quad \frac{S_{n,m}}{S_n} \rightarrow mp^{m-1}e^{-2\lambda}. \quad (8)$$

Bassan, Donderwinkel, and Kolesnik give a shorter proof of the leading result [3]. We do not present the leading enumeration problem as an open conjecture solved here.

Uniform renewal expansions and geometric-sum approximations have an established literature, including Blanchet and Glynn [4]. Olvera-Cravioto, Blanchet, and Glynn [5] explicitly study a logarithmic heavy-traffic/heavy-tail transition. Their queueing and survival-probability setting differs from the local lattice coefficient problem here. We credit the transition mechanism to that literature and prove the needed coefficient estimates directly, rather than importing an unchecked theorem. The contribution of this report is an explicit refinement for the score-sequence model, with constants, fixed-order generators, and inverse scopes. A bounded primary-source review supports this focus; it cannot certify that no equivalent refinement exists elsewhere.

2 An exact analytic and singular splitting

Throughout, let

$$\rho = \frac{1}{4}, \quad \lambda = A(\rho), \quad \mu = \rho A'(\rho), \quad p = 1 - e^{-\lambda}, \quad c = e^{-\lambda}\mu, \quad u_c = \frac{1}{p}, \quad a_c = \frac{c}{p}. \quad (9)$$

All these constants except ρ are derived from the exact series, and λ, μ, p, c, a_c are positive.

Separate the divisor $d = n$ in (2):

$$A(z) = F(z) + R(z), \quad F(z) = \frac{1}{2} \sum_{n \geq 1} \binom{2n}{n} \frac{z^n}{n^2}. \quad (10)$$

Writing $r_n = [z^n]R(z)$ gives the exact rational formula

$$r_n = \frac{2nN_n - \binom{2n}{n}}{2n^2}. \quad (11)$$

For a proper divisor d of n , $d \leq n/2$. Since $\varphi(n/d) \leq n$ and $\binom{2d}{d} \leq 4^d \leq 2^n$, the crude bound using at most n divisors gives $|r_n| \leq 2^{n-1}$. Therefore R is analytic in $|z| < 1/2$. Analyticity on the boundary $|z| = 1/2$ is not needed or asserted.

Put $t = \sqrt{1-4z}$, choosing the principal branch near the real interval $z < \rho$, and set $v = t^2$. Differentiation of the central-binomial series yields

$$\frac{dF}{dt} = -\frac{2t}{1-t^2} \log \frac{2}{1+t}. \quad (12)$$

Integrating the even and odd parts, with $F(\rho) = \pi^2/12 - \log^2 2$, gives the convergent local decomposition

$$A(z) = E(t^2) + B(t), \quad (13)$$

$$E(v) = \frac{\pi^2}{12} - \log^2 2 + \log 2 \log(1-v) - \frac{1}{4} \log^2(1-v) + R\left(\frac{1-v}{4}\right), \quad (14)$$

$$B(t) = \sum_{j \geq 1} b_j t^{2j+1}, \quad b_j = \frac{2}{2j+1} \sum_{k=0}^{j-1} \frac{1}{2k+1}. \quad (15)$$

For example, $b_1 = 2/3$, $b_2 = 8/15$, and $b_3 = 46/105$. Write $E(v) = \sum e_j v^j$. Then

$$e_0 = \lambda, \quad e_j = -\frac{\log 2}{j} - \frac{H_{j-1}}{2j} + \frac{(-1)^j R^{(j)}(\rho)}{4^j j!} \quad (j \geq 1), \quad (16)$$

where $H_0 = 0$. In particular $e_1 = -\mu$ and

$$e_2 = -\frac{\log 2}{2} - \frac{1}{4} + \frac{R''(\rho)}{32}. \quad (17)$$

Thus, with $w = 1 - 4z$,

$$A(z) = \lambda - \mu w + \frac{2}{3} w^{3/2} + e_2 w^2 + O(w^{5/2}), \quad (18)$$

$$I(z) = p - cw + \frac{2}{3} e^{-\lambda} w^{3/2} + O(w^2). \quad (19)$$

These are analytic Puiseux expansions on a fixed slit neighborhood, not merely real-axis estimates.

The split also accelerates computation:

$$\lambda = \frac{\pi^2}{12} - \log^2 2 + \sum_{n \geq 1} \frac{r_n}{4^n}, \quad \mu = \log 2 + \sum_{n \geq 1} \frac{nr_n}{4^n}. \quad (20)$$

After terms through M , the absolute tail bounds are respectively 2^{-M-1} and $(M+2)2^{-M-1}$. Derivative tails are similarly bounded by a polynomial times 2^{-M} . Exact-integer subtraction in (11) avoids cancellation before the conversion to floating point.

Constant	Approximate value	Constant	Approximate value
λ	0.3302375439859293	μ	0.6739526018515949
e_2	-0.5989545674814332	u_c	3.5555932000209863
a_c	1.7223486864283872	$k = 2/(3\mu)$	0.9891892469

The displayed decimals are numerical evaluations, not certified enclosures.

2.1 The renewal interpretation

Define $q_j = I_j 4^{-j}/p$ for $j \geq 1$. This is a probability mass function, and its generating function is $I(y/4)/p$. Its mean is $\sum j q_j = \rho I'(\rho)/p = c/p = a_c$. Equation (8) gives a mass tail proportional to $j^{-5/2}$, hence finite mean and infinite variance. At $u = u_c$, $W(y/4, u_c)$ is the renewal generating function. The limiting constant a_c^{-1} is the classical finite-mean renewal behavior. This interpretation motivates the two mechanisms but is not used in place of the contour proof.

3 Fixed weights and explicit enumeration corrections

For $\sigma \in \{+1, -1\}$, define $T_n^+ = S_n$ and $T_n^- = I_n$ for $n \geq 1$. Introduce

$$D_j^\sigma = [t^{2j+3}] \exp(\sigma(E(t^2) - \lambda)) \sinh B(t), \quad j \geq 0. \quad (21)$$

Both signs use the same $\sinh B$: the odd part of $\exp(\sigma B)$ is $\sigma \sinh B$, and the minus sign in $-\exp(-A)$ cancels it for I_n .

Define the standard gamma-ratio coefficients by

$$\frac{\Gamma(n - \alpha)}{\Gamma(n + 1)} \sim n^{-\alpha-1} \sum_{\ell \geq 0} g_\ell(\alpha) n^{-\ell}. \quad (22)$$

They result by exponentiating the formal series

$$\sum_{j \geq 1} \frac{(-1)^{j+1} (B_{j+1}(-\alpha) - B_{j+1}(1))}{j(j+1)n^j}, \quad (23)$$

where $B_j(x)$ is the Bernoulli polynomial. In particular $g_0 = 1$ and $g_1(\alpha) = \alpha(\alpha + 1)/2$.

Theorem 3.1 (All fixed orders for all and strong scores). *For every fixed integer $K \geq 0$,*

$$T_n^\sigma = \frac{e^{\sigma\lambda}}{2\sqrt{\pi}} 4^n n^{-5/2} \left(\sum_{m=0}^K \frac{c_m^\sigma}{n^m} + O_K(n^{-K-1}) \right), \quad (24)$$

where

$$c_m^\sigma = 2\sqrt{\pi} \sum_{j=0}^m \frac{D_j^\sigma g_{m-j}(j + 3/2)}{\Gamma(-j - 3/2)}. \quad (25)$$

The first three coefficients are

$$\begin{aligned} c_0^\sigma &= 1, \\ c_1^\sigma &= -\frac{1}{8} + \sigma \frac{5}{2} \mu, \\ c_2^\sigma &= \frac{1}{128} + \frac{35}{8} \mu^2 + \sigma \left(\frac{63}{16} \mu + \frac{35}{4} e_2 \right). \end{aligned} \quad (26)$$

Every finite collection is computable from finitely many derivatives of R at ρ .

Proof. The exact split continues to a dented neighborhood of $z = \rho$, while F is analytic in the plane slit along $[\rho, \infty)$ and R is analytic for $|z| < 1/2$. Thus ρ is the only singularity on the convergence

circle. The odd local part of either generating function is $e^{\sigma\lambda} \sum_{j \geq 0} D_j^\sigma t^{2j+3}$. For noninteger α the exact coefficient identity is

$$[z^n](1-4z)^\alpha = 4^n \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}. \quad (27)$$

Apply (22) to each retained odd power and collect equal relative powers of n^{-1} ; this gives (25).

For the error, retain additional terms of the convergent Puiseux expansion beyond the requested order and use the same dented coefficient contour. The next nonanalytic term and its gamma-ratio expansion give $O(4^n n^{-K-7/2})$ after terms through relative order K . The locally analytic even part is handled by analytic continuation of the full expansion and contour deformation. One must not infer an exponentially small error from a merely local Taylor polynomial, or apply an unqualified transfer rule to an integer-power remainder. Keeping enough further Puiseux terms and the common continuation domain avoids both errors.

Finally $D_0 = 2/3$ and $D_1^\sigma = 8/15 - (2/3)\sigma\mu$. Expanding one term further and using (23) yields (26). $\square$

The second corrections are approximately $c_2^+ = -0.5921736164232970$ and $c_2^- = 4.5821545749204735$. This is a fixed-order asymptotic expansion; it makes no assertion that the infinite series in $1/n$ converges.

Proposition 3.2 (Fixed subcritical and critical weights). *For fixed $0 < u < u_c$, uniformly on compact subsets of that interval,*

$$W_n(u) \sim \frac{ue^{-\lambda}}{(1-up)^2} \frac{4^n}{2\sqrt{\pi}n^{5/2}}. \quad (28)$$

There is a full expansion in relative integer powers of $1/n$ at every fixed order, generated by the odd t coefficients of

$$\frac{1}{1-u+u\exp\{-E(t^2)-B(t)\}}$$

and (27). At the critical weight,

$$4^{-n}W_n(u_c) = \frac{1}{a_c} + \frac{2}{3\mu a_c \sqrt{\pi}} n^{-1/2} + O(n^{-3/2}). \quad (29)$$

There is no n^{-1} term.

Proof. For subcritical u , $|uI(z)| \leq up < 1$ on $|z| \leq \rho$, so the analytic composition has no zero denominator there. Compactness and the continuation in Section 2 provide a common dented neighborhood on compact weight sets. The first odd term is $\frac{2}{3}ue^{-\lambda}(1-up)^{-2}t^3$; transfer gives (28). At u_c , Laurent expansion gives

$$W(z, u_c) = a_c^{-1}t^{-2} + \frac{2}{3\mu a_c}t^{-1} + O(1).$$

The exact t^{-2} coefficient is 4^n . The next nonanalytic term after t^{-1} is t , of order $4^n n^{-3/2}$. Nonnegative even powers are locally analytic and produce no algebraic n^{-1} term under the full continuation argument. $\square$

4 A uniform logarithmic near critical expansion

Let

$$u_n(\tau) = \frac{n}{np + \tau c}, \quad a_n(\tau) = u_n(\tau)c, \quad k = \frac{2}{3\mu}. \quad (30)$$

Thus $0 \leq \tau$ corresponds to $u_n(\tau) \leq u_c$. We sometimes write u, a for $u_n(\tau), a_n(\tau)$.

Theorem 4.1 (First two uniform terms). *Fix $C > 0$. Uniformly for $0 \leq \tau \leq C \log n$,*

$$4^{-n} W_n(u_n(\tau)) = \frac{1}{a_n(\tau)} \left\{ e^{-\tau} + kn^{-1/2} H(\tau) \right\} + O_C(n^{-1}), \quad (31)$$

where the entire function

$$H(\tau) = \frac{1}{\sqrt{\pi}} {}_1F_1(2; 1/2; -\tau) = \sum_{j \geq 0} \frac{(-1)^j (j+1) \tau^j}{\Gamma(j+1/2)}. \quad (32)$$

The remainder is absolute on the 4^{-n} normalization. The displayed model is eventually positive throughout this window, and its relative error is $O_C((\log n)^2 n^{-1/2})$. Replacing $e^{-\tau}$ by the exact carrier $(1 + \tau/n)^{-n-1}$ preserves the absolute bound.

4.1 One common coefficient contour

Use $y = 4z$, $w = 1 - y$, and $q = \tau/n$. By (19) and (30),

$$1 - uI(y/4) = a\{w + q - kw^{3/2} + O(w^2)\}. \quad (33)$$

The remainder constant is uniform for u in a compact positive neighborhood of u_c , which contains the whole parameter window for large n .

Fix $\pi/2 < \theta < \pi$. On the rays $w = re^{\pm i\theta}$ and throughout the joining arc $|\arg w| \leq \theta$,

$$|w + q| \geq \cos(\theta/2)(|w| + q), \quad q \geq 0. \quad (34)$$

The perturbation in (33) is therefore less than half the leading denominator in a single sufficiently small neighborhood, independent of q . This excludes local zeros on the indented sector. At $q = 0$, the critical endpoint singularity is encircled by the indentation, not omitted.

Away from $y = 1$ on $|y| = 1$, positivity and aperiodicity give $|I(y/4)| < p$. Indeed, equality would require the positive terms at degrees 1, 3, 4 to have the same phase; $y = y^3 = y^4$ forces $y = 1$. A compact arc away from 1 has a strict nonvanishing margin. Analytic continuation from $F + R$ thickens this arc slightly while retaining that margin uniformly as $u \rightarrow u_c$.

Combining this compact continuation with the local sector yields a common dented contour: two local rays joined to an indentation $|w| = 1/n$, and outer parts with $|y| \geq 1 + \delta$ for some fixed $\delta > 0$ away from the local pieces. Outer contributions are exponentially small. This argument asserts a suitable zero-free contour neighborhood, not an unjustified zero-free disk beyond the original circle.

4.2 Resolvent expansion and its error

On the local contour, a geometric-resolvent expansion gives

$$W(y/4, u) = \frac{1}{a(w+q)} + \frac{k}{a} \frac{w^{3/2}}{(w+q)^2} + E_n(w), \quad (35)$$

where

$$|E_n(w)| \leq K \left\{ \frac{|w|^2}{(|w| + q)^2} + \frac{|w|^3}{(|w| + q)^3} \right\} \leq 2K. \quad (36)$$

For sufficiently small ray radius, $|(1 - w)^{-n-1}| \leq K \exp(-cn|w|)$. Thus the ray integral of (36) is $O(n^{-1})$; the indentation has length $O(n^{-1})$ and bounded coefficient kernel. The common outer contour adds only exponentially small terms.

The first model coefficient is exact:

$$[y^n](w + q)^{-1} = (1 + q)^{-n-1}. \quad (37)$$

Its difference from $e^{-\tau}$ is $O(n^{-1})$ uniformly on the stated logarithmic window: the exponential factor absorbs the polynomial in τ from Taylor expansion.

For the fractional model set $s = nw$. Let $\mathcal{H}$ be the fixed-angle Hankel contour starting at infinity on the lower ray, following the unit right-hand arc from angle $-\theta$ to θ , and returning to infinity on the upper ray. The principal power of s is used. On this contour, $|s + \tau| \geq c(|s| + \tau)$ and $|s^{3/2}/(s + \tau)^2| \leq K|s|^{-1/2}$. Hence

$$\begin{aligned} [y^n] \frac{w^{3/2}}{(w + q)^2} &= n^{-1/2} \frac{1}{2\pi i} \int_{\mathcal{H}} e^s \frac{s^{3/2}}{(s + \tau)^2} ds + O(n^{-3/2}) \\ &= n^{-1/2} H(\tau) + O(n^{-3/2}). \end{aligned} \quad (38)$$

To justify replacing $(1 - s/n)^{-n-1}$ by e^s , first restrict to $|s| \leq n^\eta$ with fixed $\eta < 1/2$. The difference is bounded by $O((|s| + |s|^2)/n)$ times a decaying exponential. This is uniformly integrable against the preceding bound. The remaining rays are exponentially suppressed. For $|\tau| < 1$, expanding $(s + \tau)^{-2}$ on the unit indentation and using the reciprocal-gamma Hankel formula gives (32); analytic continuation extends the identity along $\tau \geq 0$. The model pole at $s = -\tau$ is on the excluded cut, so no residue has been silently removed. Equations (35)–(38) prove the absolute part of Theorem 4.1.

4.3 Positivity and relative error

Repeated expansion of the inverse-Laplace integral gives

$$H(\tau) = \frac{3}{4\sqrt{\pi}\tau^2} \left(1 + \frac{5}{\tau} + \frac{105}{4\tau^2} + O(\tau^{-3}) \right) \quad (\tau \rightarrow +\infty). \quad (39)$$

For instance expand $(s + \tau)^{-2}$ on $|s| \leq \tau/2$ and bound the rest of the fixed rays exponentially. Reciprocal gamma values give the displayed coefficients. In particular $H(\tau) > 0$ eventually and $H(\tau) \asymp \tau^{-2}$ there. On a fixed initial interval, $e^{-\tau}$ dominates the bounded $n^{-1/2}H(\tau)$ term; no global positivity of H alone is required. The full model is consequently bounded below by a positive constant times $n^{-1/2}/(\log n)^2$ on the window. Dividing the absolute error by this bound proves the relative statement.

5 The half power generator to every fixed order

The exact local denominator series is

$$D(w) = \frac{p - I((1 - w)/4)}{c} = \sum_{j \geq 2} d_j w^{j/2}, \quad d_2 = 1, \quad d_3 = -k, \quad d_4 = \ell = \frac{\mu^2/2 - e_2}{\mu}. \quad (40)$$

In particular $\ell \simeq 1.225695427220266$. The sign before e_2 is minus, because $p - I = \exp(-A) - \exp(-\lambda)$.

5.1 Entire kernel functions

For the finite monomials arising below, define

$$\mathcal{K}_{\alpha,b}(\tau) = \frac{1}{2\pi i} \int_{\mathcal{H}} e^s \frac{s^\alpha}{(s+\tau)^b} ds = \sum_{j \geq 0} \frac{(-1)^j (b)_j \tau^j}{j! \Gamma(b+j-\alpha)}. \quad (41)$$

The contour identity holds for $\tau \geq 0$, and the convergent series supplies the entire continuation. Here $(b)_j$ denotes the rising factorial and $1/\Gamma$ is its entire reciprocal, including its zeros. Equivalently this is ${}_1\tilde{F}_1(b; b-\alpha; -\tau)$, the *regularized* confluent hypergeometric function. If $b-\alpha$ is a nonpositive integer, the unregularized product ${}_1F_1(b; b-\alpha; -\tau)/\Gamma(b-\alpha)$ is not a literal defined expression. Formula (41) avoids this degeneracy.

Theorem 5.1 (Uniform expansion at every fixed order). *Put $\varepsilon = n^{-1/2}$ and define $C_m(\tau)$ by the finite formal coefficient extraction*

$$C_m(\tau) = \frac{1}{2\pi i} \int_{\mathcal{H}} e^{s[\varepsilon^m]} \frac{\exp \left\{ \sum_{j \geq 1} \varepsilon^{2j} \left(\frac{s^j}{j} + \frac{s^{j+1}}{j+1} \right) \right\}}{s + \tau + \sum_{j \geq 1} d_{j+2} \varepsilon^j s^{1+j/2}} ds. \quad (42)$$

For each fixed m , this is a finite linear combination of the entire kernels in (41). For every fixed integer $M \geq 0$ and fixed $C > 0$,

$$4^{-n} W_n(u_n(\tau)) = \frac{1}{a_n(\tau)} \sum_{m=0}^M n^{-m/2} C_m(\tau) + O_{M,C}(n^{-(M+1)/2}) \quad (43)$$

uniformly on $0 \leq \tau \leq C \log n$. In particular $C_0 = e^{-\tau}$, $C_1 = kH(\tau)$, and

$$C_2(\tau) = e^{-\tau} \left\{ \frac{\tau^2}{2} - \tau - \ell(\tau^2 - 2\tau) + k^2 \left(-3\tau + 3\tau^2 - \frac{\tau^3}{2} \right) \right\}. \quad (44)$$

For $M \geq 1$ the model is eventually positive, and its relative error is $O_{M,C}((\log n)^2 n^{-M/2})$.

Proof. On the contour of Section 4.1, write $D(w) = w + V(w)$ with $V(w) = O(w^{3/2})$. Inequality (34) gives $V(w)/(w+q) = O(|w|^{1/2})$ uniformly in $q \geq 0$. Expanding the resolvent through M such factors leaves a remainder bounded by

$$K \frac{|w|^{(M+1)/2}}{|w|+q} \leq K |w|^{(M-1)/2}. \quad (45)$$

Since the local Puiseux series is genuinely convergent, truncating the numerators at the corresponding total half-power order has the same or a smaller error. Integration along the rays and the radius-1/ n indentation makes (45) an $O(n^{-(M+1)/2})$ coefficient error, including $M = 0$.

After $s = nw$, the exact coefficient kernel has the identity

$$(1 - \varepsilon^2 s)^{-\varepsilon^{-2}-1} = e^s \exp \left\{ \sum_{j \geq 1} \varepsilon^{2j} \left(\frac{s^j}{j} + \frac{s^{j+1}}{j+1} \right) \right\}. \quad (46)$$

For the prescribed fixed M , choose a sufficiently small fixed $\eta > 0$ and expand on $|s| \leq n^\eta$. Taylor remainders are bounded by the required power of $n^{-1/2}$ times a fixed polynomial in $|s|$ and a decaying exponential. The denominator bound gives uniform integrable domination for each finite coefficient and remainder. Outside this range both the exact kernel and the finitely many model terms are exponentially small. The common outer contour also contributes exponentially small terms. This proves (42) and (43), with constants depending on M, C but not on τ, n .

At order ε^2 , the three kernels are

$$\frac{s + s^2/2}{s + \tau}, \quad -\ell \frac{s^2}{(s + \tau)^2}, \quad k^2 \frac{s^3}{(s + \tau)^3}.$$

Their time-one inverse Laplace values yield (44). Polynomial parts in s have zero Hankel integral (or are distributions at time zero), so they do not contribute at time one. In particular $C_2(0) = 0$, in agreement with (29).

Each fixed C_m is bounded on $\tau \geq 0$ by its fixed-contour integral. The extra terms for $m \geq 2$ therefore total $O(n^{-1})$, which is negligible compared with the positive lower bound from Theorem 4.1. Positivity and the relative error follow. $\square$

The quantifiers are essential: choose the truncation order and logarithmic window constant *before* taking $n \rightarrow \infty$. The theorem does not claim uniformity in growing M , convergence of its infinite formal expansion, or validity for a supercritical weight.

6 The explicit two contribution coexistence profile

Set

$$b_n = \frac{1}{2} \log n + 2 \log \log n + \log \frac{\mu \sqrt{\pi}}{2}, \quad \tau_n(s) = b_n + s. \quad (47)$$

For s in a fixed compact real interval, this parameter is nonnegative and lies in a fixed logarithmic window for sufficiently large n .

Corollary 6.1 (Coexistence). *Uniformly for s in compact real intervals,*

$$a_c \sqrt{n} (\log n)^2 4^{-n} W_n(u_n(\tau_n(s))) \longrightarrow \frac{2}{\mu \sqrt{\pi}} (1 + e^{-s}). \quad (48)$$

Proof. The exponential term satisfies $\sqrt{n} (\log n)^2 e^{-b_n - s} = 2e^{-s}/(\mu \sqrt{\pi})$. For the second term, (39) and $\tau_n(s)/\log n \rightarrow 1/2$ give $k(\log n)^2 H(\tau_n(s)) \rightarrow 2/(\mu \sqrt{\pi})$. Also $a_n(\tau_n(s)) \rightarrow a_c$. The normalized absolute error in Theorem 4.1, multiplied by $\sqrt{n} (\log n)^2$, tends to zero. $\square$

The balance is between the near-critical exponential coefficient contribution and the heavy-tail correction. The $2 \log \log n$ term is necessary because the heavy-tail contribution contains τ^{-2} . A compact τ window cannot see this coexistence balance. Logarithmic heavy-traffic/heavy-tail

transitions are established phenomena [5]; (48) supplies the model-specific centering and local-coefficient profile.

Nothing in the coefficient decomposition alone identifies the conditional law of the number or sizes of blocks in either contribution. In particular, (48) is not a proved distributional mixture, and the factors 1 and e^{-s} are not asserted to be mixture probabilities.

[write] **Added 3 October 2026, batch 85.** Part II supplies the separate uniform estimates that this paragraph says the coefficient decomposition alone cannot give. It computes the weight of the event that one block exceeds $n/2$ exactly (101), shows that this weight is asymptotic to the heavy-tail term ((102), by the giant-deletion Theorem 14.2) and that the complementary event carries the exponential term ((112), exhausted by local-limit windows). Hence, at $\tau = b_n + s$, the factor 1 in (48) is the limiting relative weight of the giant-block event and the factor e^{-s} that of the event with a linear number of blocks, and $1/(1 + e^{-s})$ and $e^{-s}/(1 + e^{-s})$ are mixture probabilities (Theorem 12.1). The two sentences above remain true of (48) on its own. Part II is unrefereed, prepared with AI assistance and not formalized.

7 What can be inverted and with what error

An inverse requires a specified one-variable map. We give three such choices. In every case, a real inverse-location error is distinct from a guarantee about rounding to an integer threshold.

7.1 Fixed family size from an all or strong count

Let $h = \log 4$ and $C_\sigma = e^{\sigma\lambda}/(2\sqrt{\pi})$. For fixed K , define the eventually positive model

$$F_{K,\sigma}(x) = C_\sigma 4^x x^{-5/2} \left(1 + \sum_{j=1}^K c_j^\sigma x^{-j} \right). \quad (49)$$

It is eventually strictly increasing, with logarithmic derivative $h + O(1/x)$. Denote its eventual real inverse by $x_{K,\sigma}(Y)$. Theorem 3.1 and the mean value theorem then imply, whenever $Y = T_n^\sigma$ and n is sufficiently large,

$$n = x_{K,\sigma}(Y) + O(x_{K,\sigma}(Y)^{-K-1}). \quad (50)$$

For $K = 0$ the model inverse is explicit:

$$x_{0,\sigma}(Y) = -\frac{5}{2h} W_{-1} \left(-\frac{2h}{5} \left(\frac{C_\sigma}{Y} \right)^{2/5} \right). \quad (51)$$

The branch W_{-1} selects the large positive solution. Higher-order inverses can be found by substitution or Newton iteration on $\log F_{K,\sigma}(x) - \log Y$.

7.2 Size along a specified coexistence path

Fix $s \in \mathbb{R}$. Extend (47) to large real x and define

$$G_s(x) = \frac{4^x}{a_x(\tau_x)} \left\{ e^{-\tau_x} + kx^{-1/2} H(\tau_x) \right\}, \quad \tau_x = \frac{1}{2} \log x + 2 \log \log x + \log \frac{\mu\sqrt{\pi}}{2} + s. \quad (52)$$

This is eventually positive with logarithmic derivative tending to h . If $x_s(Y)$ is its eventual inverse and $Y = W_n(u_n(\tau_n(s)))$, then

$$n = x_s(Y) + O\left(\frac{(\log x_s(Y))^2}{\sqrt{x_s(Y)}}\right). \quad (53)$$

This uses the explicit derivative of the model, not differentiation of an uncontrolled enumeration remainder. An elementary first inverse center, with $L = \log Y$, is

$$x = \frac{L + \frac{1}{2} \log L + 2 \log \log L - \frac{1}{2} \log h - \log K_s}{h} + o(1), \quad K_s = \frac{2(1 + e^{-s})}{a_c \mu \sqrt{\pi}}. \quad (54)$$

The discrete weighted sequence along this fixed path is eventually increasing because its consecutive ratio tends to 4. An exact threshold still requires bracketing: small inverse error does not resolve an integer boundary without a certified separation.

7.3 Known size and an unknown component weight

Fix a compact real interval J . Suppose n is known, $\tau = b_n + s$ with $s \in J$, and the target is specified as

$$Y = 4^{-n} W_n(u_n(\tau)). \quad (55)$$

For $n \geq 1$, $W_n(u)$ is a nonconstant polynomial with nonnegative coefficients and is strictly increasing on $u > 0$. Since $u_n(\tau)$ decreases strictly, the exact target determines τ uniquely on the specified window.

For fixed $M \geq 1$, define

$$\mathcal{M}_{n,M}(t) = \left(\frac{p}{c} + \frac{t}{n}\right) \sum_{m=0}^M n^{-m/2} C_m(t). \quad (56)$$

For all sufficiently large n , this model is strictly decreasing on the padded interval $b_n + (J + [-1, 1])$ and has derivative magnitude at least a positive constant times $n^{-1/2}/(\log n)^2$. Indeed, the C_0 derivative is negative of that size, the C_1 derivative is smaller by $O(1/\log n)$ using $H'(t) = O(t^{-3})$, and the higher derivatives are bounded by their contour integrals. The derivative $1/n$ of the prefactor is negligible. Only the explicit model is differentiated.

The padding ensures the model brackets the exact target even when s is at an endpoint of J : the model change over a unit of padding dominates the absolute error in (43). There is therefore one root $\hat{\tau}$ of $\mathcal{M}_{n,M}(\hat{\tau}) = Y$ on that padded interval, and the mean value theorem gives

$$|\hat{\tau} - \tau| = O_{M,J}((\log n)^2 n^{-M/2}). \quad (57)$$

Since $|u'_n(t)| = nc/(np + tc)^2 = O(n^{-1})$ there, the inferred weight $\hat{u} = n/(np + \hat{\tau}c)$ satisfies

$$|\hat{u} - u_n(\tau)| = O_{M,J}((\log n)^2 n^{-1-M/2}). \quad (58)$$

The displayed C_2 thus improves weight localization from $O(\log^2 n/n^{3/2})$ to $O(\log^2 n/n^2)$. These are asymptotic localization bounds for a known- n model, not finite- n certified error bars or an unspecified inverse of a bivariate sequence.

[write] **Added 2 October 2026, batch 77.** The general steps of this section are instances of results of the repository's transseries volume ([Analysis/Transseries/docs/series-and-transseries/](#)

`Transseries_And_Inversion/`), and no novelty is claimed for them. The explicit inverse (51) is `p0:thm:lambert-core` for $hx - \frac{5}{2} \log x = \log(Y/C_\sigma)$, that is $a = h = \log 4$, $b = -5/2$, with the branch W_{-1} that its branch rule selects for the large solution. The three localization statements (50), (53) and (57) divide a model error by a model slope, which is the residual-to-root conversion `p0:thm:backward-error`. The remark that an exact threshold still needs bracketing is the separation condition, part (2) of `p0:thm:staircase`; no exact-rounding rule (its part (1)) is used. The generic staircase arithmetic is formalized in Lean, in the file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, as `Fabius.staircase_ceil` and `Fabius.staircase_separation`; those lemmas concern an arbitrary monotone function, not S_n or W_n , and give this report no formal status.

8 Reproducible checks and numerical interpretation

The accompanying archive contains this editable T_EX source, the compiled PDF, a portable Python verification program, generated JSON data, a replay script, and a SHA-256 manifest. It does not require a connection to OEIS or another external data service to run the mathematical checks. The code independently generates the integer recurrences rather than loading an OEIS term list.

The check layers have different evidentiary roles:

1. **Exact enumeration.** Enumerate all nondecreasing candidates through a small size, apply Landau's inequalities, and count equality indices. Compare the resulting $S_n, I_n, S_{n,m}$ with the divisor and renewal recurrences. Integer-weight evaluations test the component polynomial directly.
2. **Exact algebra.** Verify the Puiseux and gamma-ratio formulas for c_1^σ, c_2^σ , the denominator coefficient $d_4 = (\mu^2/2 - e_2)/\mu$, and the three-kernel formula for C_2 . Such assertions are algebraic identity checks.
3. **Finite generator.** Construct the formal half-power coefficients through $M = 4$, evaluating integer-exponent kernels by polynomial/rational decomposition and nondegenerate half-integer kernels by Kummer functions. This avoids undefined ordinary hypergeometric parameters.
4. **Numerical recurrence and inverse comparisons.** Compute normalized weighted coefficients, compare fixed-order models, and solve the explicit known- n inverse. These finite checks can detect implementation mistakes; they do not prove uniform asymptotics, certify a finite- n bound, or establish novelty.

For illustration, at the coexistence center $\tau = b_n$, let $Q = 4^{-n}W_n(u_n(b_n))$. The following values use the corrected C_2 and the relative-error convention $Q/\mathcal{M}_{n,M}(b_n) - 1$:

n	b_n	Q	$M = 1$ error	$M = 2$ error
100	4.841567	0.00618023445	-6.722%	1.585%
300	5.818773	0.00261743283	-5.724%	0.792%
1000	6.803789	0.00105627898	-3.461%	0.235%

These examples illustrate a finite-size benefit of the extra term, not a universal monotonic-error claim. At $\tau = 0$, $C_2(0) = 0$, so the $M = 1$ and $M = 2$ models coincide. Higher generator terms are needed to improve that special point.

For the fixed families, the residual $n^2(T_n^\sigma/(C_\sigma 4^n n^{-5/2}) - 1 - c_1^\sigma/n)$ approaches c_2^σ . Representative values are

n	All-score residual	Strong-score residual
100	−0.6704789062	4.4569099700
200	−0.6311256921	4.5186728913
1000	−0.5999299576	4.5693162876
Limit coefficient	−0.5921736164	4.5821545749

All decimal computations, including values computed at high working precision, remain non-interval checks. The proof of each remainder is analytical. A validated finite- n certificate would require interval evaluation and explicit constants throughout the contour estimates.

8.1 How to replay

From the extracted archive, run `bash replay.sh`. The script runs exact and symbolic assertions, regenerates numerical JSON data, and invokes `pdflatex` twice to rebuild the PDF. Python requirements are listed in `requirements.txt`; a working L^AT_EX installation is also needed. Use `python3 scripts/verify_manifest.py` before replay to check the packaged file hashes. The README specifies versions and commands. Generated results are written inside the extracted directory, and the source code has no absolute workspace paths.

[write] **Added 2 October 2026, batch 77.** This subsection describes the delivered archive. In the repository the files keep their bytes but not their places: `scripts/` is `code/`, `results/` is `data/`, `replay.sh` and `build_pdf.sh` are in `code/`, `requirements.txt` is in `data/`, and `report.tex` became `article.tex`. The delivered PDF and the manifest SHA256SUMS were verified at placement and not shipped (both survive in the arrival commit), so `verify_manifest.py` cannot run as shipped. `run_checks.py` writes into `results/` beside `scripts/`, overwriting the recorded outputs; the report’s README.md therefore runs it on a scratch copy with the delivered layout. At intake (2 October 2026) that run reported status PASS and all seven JSON files equal to the recorded ones up to line endings. An independent intake computation (exact S_n to $n = 1200$ from the totient formula) gave $\lambda = 0.33023754398592927\dots$ and $\mu = 0.67395260185159494\dots$, $n(S_n/(C_+ 4^n n^{-5/2}) - 1) = 1.55938$ at $n = 1200$ against $c_1^+ = 1.55988$, and second-order residuals -0.59993 ($n = 1000$) and -0.59864 ($n = 1200$) approaching $c_2^+ = -0.59217$ at rate $1/n$, consistent with Theorem 3.1.

9 Further questions

The results suggest several concrete extensions without settling them:

1. **Conditional component laws at coexistence.** Identify which component-count and component-size ranges contribute to the two coefficient terms. A genuine mixture theorem needs separate uniform estimates for those ranges, rather than only the sum of their weights.
2. **Supercritical crossover.** For $\tau < 0$, the denominator can develop a moving positive-real pole. Its residue and interaction with the branch point require a revised contour analysis; the present zero-exclusion proof does not apply unchanged.
3. **Wider parameter windows and explicit constants.** Determine the maximal useful growth of τ for a chosen truncation and turn the asymptotic O constants into practical, interval-certified bounds. This would support reliable finite-size thresholds and certified weight inversion.
4. **Growing component counts or truncation orders.** Fixed- m laws and fixed- M expansions

do not cover m or M growing with n . Uniformity in those directions may involve additional scales and optimal truncation questions.

5. **Comparison with general renewal theorems.** Locate the strongest existing local lattice theorem that specializes to the present half-power structure, and compare its hypotheses and remainder constants with the explicit generator. This is necessary before claiming priority for any general mechanism.

[write] **Added 3 October 2026, batch 85.** Question 1 is answered, for the weight $u_n(b_n + s)$ with s in a compact interval, by Part II (unrefereed, AI-assisted, not formalized): the heavy-tail contribution comes from configurations with a unique block of size $n - O_{\mathbb{P}}(n/\log n)$, around which the remaining blocks form two nearly independent geometric renewal strings, and the exponential contribution from configurations with $n/a_c + O_{\mathbb{P}}(n^{2/3})$ blocks and largest block of order $n^{2/3}$ (Theorems 12.1–12.4). Growing shifts $s_n \rightarrow \pm\infty$ are not covered (Part II’s question RQ5). Question 2 is restated, still open, as Part II’s RQ6. For question 5, Part II compares its mechanism with the Gibbs-partition, balls-in-boxes and big-jump literature (Section 10); that comparison is a bounded source check, so the question stays open. Questions 3 and 4 are unchanged: Part II’s block counts grow with n , but under the moving weight $u_n(b_n + s)$, not in the fixed- m law of (8) or the fixed- M expansions.

Part II

Conditional phase coexistence: giant blocks, Gamma laws, and stable fluctuations

Summary. We resolve the conditional component-law question left open in Part I, the ProveIt report on tournament score sequences, OEIS A000571 and A351822. When each irreducible block receives a weight approaching its critical value at a specified logarithmic rate, we prove a logistic mixture of two structurally different regimes. One has a unique block of size $n - O_{\mathbb{P}}(n/\log n)$; the other has $n/m + O_{\mathbb{P}}(n^{2/3})$ blocks and no block of macroscopic size. In the first regime, removing the giant block leaves two independent geometric renewal strings up to a total variation error $O(1/\log n)$. This gives a Gamma(2, 1) law for both the rescaled block count and the giant’s deficit, a stable law at the next scale, and an explicit mixed Fréchet law for the second-largest block. A positive polynomial tilt of the residual block count improves the approximation to every fixed inverse-logarithmic order. In the second regime, the centered number of blocks converges to the negative of a spectrally positive 3/2-stable variable, jointly with an explicit Fréchet limit for the largest block.

The exact enumeration identity, the leading asymptotic counts, and the general phenomenon of condensation have established precedents. The contribution here is a proved conditional description in the size-dependent tournament model, together with an elementary total variation argument and all normalization constants. The analytic input from the repository is reproved, and the stable local limit theorem is proved directly by Fourier inversion. Exact enumeration checks and floating-point diagnostics accompany the article. No priority claim is made for general Gibbs-partition or heavy-tail mechanisms.

Main result in one display. Let K_n be the number of irreducible blocks and $M_{1,n}$ the largest block size. At the weight defined precisely in Section 12,

$$\left(\frac{mK_n}{n}, \frac{M_{1,n}}{n}\right) \xrightarrow{d} \frac{1}{1+e^{-s}} \delta_{(0,1)} + \frac{e^{-s}}{1+e^{-s}} \delta_{(1,0)}.$$

Here $m = 1.7223486864\dots$ is an explicitly defined mean block size. The first atom describes one giant block and a random sublinear number of remaining blocks; the second describes a linear number of blocks.

[write] **Added 3 October 2026, batch 85: provenance of Part II.** Part II is a single manuscript, printed in full: manuscript 04 of batch 85 of ProveIt’s incoming-reports intake, *Conditional phase coexistence for tournament score sequences: Giant blocks, Gamma laws, and stable fluctuations* (“Research manuscript: OEIS A000571 and A351822”, dated 3 October 2026; a 1,687-line source and a 25-page PDF; its PDF metadata name the author as “Research manuscript prepared with AI assistance”). It arrived as the archive `tournament_coexistence_research.zip` (wrapper directory `tournament_coexistence/`, main file `tournament_coexistence.tex`) in commit 317c1ce2e and was placed in commit 713149ded as an addition to this report. It pins the repository at commit d68b65ea064084fb121df9bb431b21d086f7be81 and continues this report’s `article.tex`; the report was not changed between the pin and the placement, so the manuscript read exactly the text printed as Part I. Its code, data and figures are shipped with the prefix `02-coex-`; its own L^AT_EX source, README, PDF and checksum list are not shipped and survive in the archive of the arrival commit. The manuscript is unrefereed and was prepared with AI assistance; its proofs are ordinary mathematical proofs, nothing in it is formalized in Lean or Rocq, and its place in this report confers no formal status.

What it answers. Part II answers the first further question of Part I (“Conditional component laws at coexistence”, Section 9) and supplies the separate estimates that the caveat at the end of Section 6 says the coefficient decomposition alone does not give: at $\tau = b_n + s$ the heavy-tail term (the factor 1 in (48)) is the weight of the event $\mathcal{C}_n$ that one block exceeds $n/2$, and the exponential term (the factor e^{-s}) is the weight of its complement $\mathcal{F}_n$ (Theorem 12.1). Dated notes at those two places of Part I say so.

Overlap with Part I, printed and credited. Proposition 13.2 and Corollary 13.3 re-prove Theorem 4.1 and Corollary 6.1 in the same normalization (a note in Section 13 gives the dictionary). The tail $q_j \sim dj^{-5/2}$ of Lemma 13.1 re-derives the mass tail of Part I’s renewal interpretation (Section 2, from (8)) with the constant d made explicit, and its aperiodicity argument is the one of Section 4.1. The constants (63)–(64) extend (9) ($m = a_c$), the centre (67) is (47), the split (80) is (10), the recurrences (124) are (5)–(7), and (126) is (20). The coefficients 1, 5, 105/4 that open Proposition 17.1 are those of (39): both expand the same heavy-tail kernel, a consistency check rather than a duplicate. Part II’s exact enumeration program is an independent reimplement of the Landau check of Part I’s `code/check_exact_counting.py`, extended to the giant-event polynomial. Everything else in Part II is new to the repository: Theorems 12.1–12.4, 14.2, Corollaries 14.3, 15.1, 17.2, the direct local limit Lemma 16.1, Propositions 17.1 and 18.1, and the point-process statements. The inversion (121) is an instance of a theorem of the repository’s transseries volume (note after it).

Editorial changes. Section k of the manuscript is Section $k + 9$ here, and its Appendices A and B are Appendices A and B; every label carries the prefix `tss:cx:`. The manuscript’s references to “the pinned ProveIt report” (its citation `Repo`, whose pin and file path this note records) now point to Part I, in its abstract (the summary above), in Section 10 and in its status table; the bibliographies are merged, so the shared entries are Part I’s and the four new ones (Białas–Bogacz–Burda–Johnston, Stuffer, Denisov–Dieker–Shneer, Armendáriz–Loulakis) are added. The two displayed figures are the shipped `02-coex-` files, and the shipped phase-profile figure, which the manuscript does not display, is added as Figure 3 with a caption taken from the delivered code README. The two theorem titles of Section 13 name the Part I results they restate, and the `[write]` notes are new. No statement, proof or number of the manuscript was altered.

[write] Added 3 October 2026, batch 85: notation of Part II. Part II keeps the manuscript’s symbols, and several of them clash with Part I’s. Within Part II every symbol has the meaning below; the right column gives Part I’s meaning, the tempting false reading. Shared notions keep one symbol: $\lambda, \mu, \rho, p, u_c, A, N_j, F(z), R(z), H(\tau), \mathcal{H}, W, W_n(u), W_{-1}, b_n, \tau, s$ (the coexistence shift), $q_j, S_n, I_n, E_n(w)$ (the resolvent error) and M (a fixed truncation order) mean the same in both Parts. The manuscript’s own symbol list is Appendix A.

symbol	meaning in Part II	not: meaning in Part I
m	mean critical block length $e^{-\lambda}\mu/p = 1.72234\dots$, which is Part I’s a_c	a number of strong blocks ($S_{n,m}$), or the index of c_m^σ, C_m
k	a block-count index ($S_{n,k} = [z^n]I^k$, sums over k)	$k = 2/(3\mu)$, which is Part II’s κ/m
κ	$2e^{-\lambda}/(3p) = d\Gamma(-3/2)$, the Puiseux and stable coefficient	(not used)
$K_n, K_{L,n}, K_{R,n}, K(x)$	numbers of blocks: in all, before and after the giant, of a sequence x	K a fixed truncation order or a generic constant
r, r_n, R_j	the normalized weight $r = up$, $r_n = (1 + m\tau_n/n)^{-1}$, and $rm = a_n(\tau)$ of Part I; $R_j = [z^j]R$	$r_n = [z^n]R$ (Part II’s R_n)
u_n	r_n/p , which is Part I’s $u_n(\tau_n)$	—
$a_n, a(k)$	the stable scales $(\kappa n/m)^{2/3}$ and $(\kappa k)^{2/3}$	$a_n(\tau) = u_n(\tau)c$, which is Part II’s rm
d	tail amplitude, $q_j \sim dj^{-5/2}$, $d = e^{-\lambda}/(2\sqrt{\pi}p)$; $d \mid j$ in (61) is a divisor	a divisor, or d_j the coefficients of $D(w)$
D_n, D	the coexistence scale (91); D its implicit-centre analogue in the proof of Proposition 18.1	$D(w), D_j^\sigma$

symbol	meaning in Part II	not: meaning in Part I
$B_n(r)$, B_n , $B(\delta)$	$B_n(r)$ the unnormalized weight of $\mathcal{C}_n$; B_n <i>without argument</i> the second-maximum scale (77), $B(\delta)$ the same scale as a function of δ (two meanings inside the manuscript, which its appendix discloses)	$B(t)$ the odd series, b_j , Bernoulli B_j
$Z_n(r)$, Z	$Z_n(r) = 4^{-n} W_n(r/p)$; Z the standard spectrally positive 3/2-stable variable (73)	—
$h_n(T)$, h	the density ratio in (96); $h > 0$ a small Chernoff parameter in Section 15	$h = \log 4$
δ , δ_n	$1 - r$, $1 - r_n \sim m\tau_n/n$	$\delta > 0$ a fixed contour margin (Section 4.1)
E , E_L , E_R , $\mathbb{E}$	exponential variables of mean one; $\mathbb{E}$ expectation	$E(v)$ the even part, e_j
G	a Gamma(2, 1) variable	$G_s(x)$ the path model (52)
L	$\log n$ in Proposition 18.1; a cutoff in the strong-law argument of Section 14	$L = \log Y$ in (54)
c_0	$\log(m/(4d))$ in Proposition 18.1	$c_0^\sigma = 1$, the leading fixed-order coefficient
$\mathcal{C}_n$, $\mathcal{F}_n$; C , C_0 , $C_1, \dots$	the events $\{M_{1,n} > n/2\}$ and $\{M_{1,n} \leq n/2\}$; generic or window constants	$C_m(\tau)$ the uniform terms, C_σ
$F(x)$	$\mathbb{P}(X \leq x)$ in Section 17 (elsewhere $F(z)$ as in Part I)	—
t	$\sqrt{1 - 4z}$ in Section 13; also a Fourier or Laplace variable, a size, and a ray parameter	the free weight parameter of $\mathcal{M}_{n,M}(t)$
z	the generating-function variable; in the proof of Proposition 13.2 also the Hankel variable nw	Part I's Hankel variable is $s = nw$
q , Q , X , S_k^X	$q = \tau/n$ inside the proof of Proposition 13.2 (as in Part I); Q the probability generating function of q_j ; X , X_i block lengths and S_k^X their partial sums	—
N_δ , N , T , $T_{L,n}$	string counts and total lengths	T_n^σ (S_n or I_n)
J_s , J_f , J	a Bernoulli variable, a Laplace exponent, a block count	J a compact interval
g , g_u	the stable density and its defective version	$g_\ell(\alpha)$ the gamma-ratio coefficients
ϕ , φ	$\phi(t) = Q(e^{it})$; φ Euler's totient as in Part I	—
ν , ν_δ , π_s	the Lévy intensity, the geometric-string law, the logistic weight $(1 + e^{-s})^{-1}$	—
$\mathcal{P}$, P_M , R_r , $L_{n,r}$, U , V	Poisson process, tilt polynomial, reference and actual residual laws, a uniform variable, the largest Poisson point	$R(z)$ is not R_r
I	an interval of count coordinates in Section 16.3 (elsewhere $I(z)$)	—
$M_{j,n}$	the j th largest block size	M a truncation order

10 Problem, provenance, and contribution

10.1 The particular open question addressed

The pinned ProveIt report *Tournament score sequences and weighted component asymptotics*, now Part I of this report, proves a two-term coefficient expansion and a coexistence profile, then explicitly leaves the conditional laws of the component counts and sizes open. Its Section 9, first research question, asks which structural ranges produce the two terms. Its Section 6 emphasizes that the two terms alone were not yet identified as probabilities of two events.

This article supplies that missing identification. The events are

$$\mathcal{C}_n = \{M_{1,n} > n/2\}, \quad \mathcal{F}_n = \{M_{1,n} \leq n/2\}.$$

The threshold $n/2$ makes the large block unique and gives an exact factorization. It does not impose an artificial asymptotic boundary: we prove that the maximum divided by n tends to 1 on $\mathcal{C}_n$ and to 0 on $\mathcal{F}_n$. Any fixed threshold strictly between zero and one therefore yields asymptotically the same classification.

The repository was inspected at commit `d68b65ea064084fb121df9bb431b21d086f7be81`. The continued report is under `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a000571-tournament-score-sequences/`, the directory of this report; its text at that commit is the text printed as Part I. The research index also records extensive work on other OEIS sequences; the choice here advances an explicit remaining question instead of repeating the classical enumeration theorem.

10.2 What belongs to the literature

The score-sequence identity used below is due to Claesson, Dukes, Franklín, and Stefánsson [1]; a shorter explanation is given by Bassan, Donderwinkel, and Kolesnik [3]. Kolesnik proved the leading all-score and strong-score asymptotics and the limiting law of the number of blocks at the unweighted value $u = 1$ [2]. Those are inputs and precedents, not open problems solved here.

There is also substantial prior work on the probability mechanism. Variable-box-number balls-in-boxes models already exhibit a competition between a condensed state and an extensive number of boxes [7]. Stuffer’s phase diagram for Gibbs partitions contains mixture regimes, stable count fluctuations, and descriptions of the remaining components after deletion of a maximum [8]. The large-deviation and conditional independence theory of Denisov–Dieker–Shneer and Armendáriz–Loulakis gives a general framework for a single large summand [9, 10]. Logarithmic transitions between geometric-sum and heavy-tail contributions occur in the queueing literature [5].

Stuffer’s mixture theorem uses fixed inner and outer coefficient sequences with matched polynomial tails. Its condensed component count has a finite limit. The balls-in-boxes studies establish the ensemble and its phase transition. Neither comparison is replaced by a claim that mixture limits or stable fluctuations were previously unknown.

The distinction relevant here is the changing outer weight $r_n^{K_n}$: $r_n \uparrow 1$ at distance of order $\log n/n$. The condensed number of blocks grows on the scale $n/\log n$, rather than staying bounded. We prove the explicit conditional laws in this window for tournament weights. The elementary renewal-string argument is presented in full, so none of the general large-deviation theorems is invoked with unchecked hypotheses. The literature comparison is a bounded source check, not a proof of publication priority.

Statement	Status in this article
Exact count identity and block decomposition	Established input, with the elementary decomposition recalled.
Leading $4^n n^{-5/2}$ asymptotics	Established result; the needed stronger local expansion is derived.
Two-term logarithmic coefficient law	Repository result (Theorem 4.1 and Corollary 6.1 of Part I), reproved in the manuscript; restated in Section 13.
Logistic probabilities of structural events	Proved here relative to the pinned report's open question.
Geometric-string total variation approximation	Proved directly, with an $O(1/\log n)$ error in this model.
Conditional Gamma, stable, and second-maximum laws	Proved here for the moving tournament weight.
General existence of condensation or stable limits	Credited to prior theory; no claim of a new general phenomenon.

This is an unrefereed research manuscript prepared with AI assistance. The proofs are ordinary mathematical proofs, not Lean or Rocq formalizations. The exact finite checks support the formulas but are not substitutes for the limiting arguments.

11 Score sequences and the exact renewal model

11.1 Irreducible blocks

A tournament is an orientation of a complete graph. A score sequence is its nondecreasing list of out-degrees. Landau's characterization, as stated in [1], says that a sequence of integers $(s_1, \dots, s_n)$ is a score sequence if and only if

$$0 \leq s_1 \leq \dots \leq s_n \leq n-1, \quad \sum_{i=1}^j s_i \geq \binom{j}{2} \quad (j < n), \quad \sum_{i=1}^n s_i = \binom{n}{2}. \quad (59)$$

It is *strong* or *irreducible* if every proper-prefix inequality is strict. A singleton is irreducible.

If equality holds at a proper prefix of length j , split the sequence there and subtract j from each entry of the second part. Both parts satisfy (59). For example, the prefix sum condition for the translated second part follows from

$$\binom{j+\ell}{2} - \binom{j}{2} - j\ell = \binom{\ell}{2}.$$

The first translated entry is nonnegative because the next prefix inequality implies $s_{j+1} \geq j$. Conversely, concatenating a score sequence of length j with one whose entries have all been increased by j preserves (59). Cutting at every equality therefore gives a unique ordered list of irreducible blocks.

Let S_n count all score sequences and I_n count nonempty strong ones, with $S_0 = 1$ and $I_0 = 0$. These are respectively A000571 and the positive-index part of A351822 [6]. The latter OEIS entry gives the empty object the value one; our generating function omits it. Write

$$S(z) = \sum_{n \geq 0} S_n z^n, \quad I(z) = \sum_{n \geq 1} I_n z^n, \quad S_{n,k} = [z^n] I(z)^k.$$

Unique decomposition gives

$$S(z) = \frac{1}{1 - I(z)}, \quad W(z, u) = \frac{1}{1 - uI(z)}, \quad W_n(u) = \sum_{k=1}^n S_{n,k} u^k \quad (n \geq 1). \quad (60)$$

The probability law studied here assigns to each score sequence x of length n the probability $u^{K(x)}/W_n(u)$. Thus the random objects are score sequences, not uniformly random labeled tournaments.

11.2 Exact enumerative input and constants

Define

$$N_j = \frac{1}{2j} \sum_{d|j} (-1)^{j+d} \varphi(j/d) \binom{2d}{d}, \quad A(z) = \sum_{j \geq 1} \frac{N_j}{j} z^j. \quad (61)$$

Here N_j is A145855. The established identity is

$$S(z) = \exp A(z), \quad I(z) = 1 - \exp(-A(z)). \quad (62)$$

We use (62) as a cited combinatorial theorem [1, 3]; we do not reprove its nontrivial bijection.

Set

$$\rho = \frac{1}{4}, \quad \lambda = A(\rho), \quad \mu = \rho A'(\rho), \quad p = 1 - e^{-\lambda}, \quad (63)$$

$$u_c = p^{-1}, \quad m = \frac{e^{-\lambda} \mu}{p}, \quad d = \frac{e^{-\lambda}}{2\sqrt{\pi} p}, \quad \kappa = \frac{2e^{-\lambda}}{3p} = d\Gamma(-3/2). \quad (64)$$

The probability generating function of a critical block length is

$$Q(y) = \frac{I(\rho y)}{p} = \sum_{j \geq 1} q_j y^j, \quad q_j = \frac{I_j 4^{-j}}{p}. \quad (65)$$

In particular, $Q(1) = 1$ and $\mathbb{E}X = Q'(1) = m$ for X with law q .

For $r = up$, the normalized weighted count is exactly

$$Z_n(r) = 4^{-n} W_n(u) = [y^n] \frac{1}{1 - rQ(y)} = \sum_{k=1}^n r^k \mathbb{P}(X_1 + \cdots + X_k = n). \quad (66)$$

For any ordered positive lengths $\ell_1 + \cdots + \ell_k = n$, their total unnormalized weight is $r^k \prod_i q_{\ell_i}$. Conditioning this measure on total length n is identical to the score-sequence law after the internal blocks have been summed out.

12 Precise statements of the conditional limits

Use the constants in (63)–(64), and put

$$b_n = \frac{1}{2} \log n + 2 \log \log n + \log \frac{m}{4d} = \frac{1}{2} \log n + 2 \log \log n + \log \frac{\mu \sqrt{\pi}}{2}. \quad (67)$$

Fix $s \in \mathbb{R}$ and, for all sufficiently large n , let

$$\tau_n = b_n + s, \quad r_n = \frac{1}{1 + m\tau_n/n}, \quad u_n = \frac{r_n}{p}, \quad \delta_n = 1 - r_n = \frac{m\tau_n}{n + m\tau_n}. \quad (68)$$

The probability law below always uses the weight u_n . Let $M_{1,n} \geq M_{2,n} \geq \dots$ be the ranked block sizes; missing sizes are zero. On $\mathcal{C}_n$, let $K_{L,n}, K_{R,n}$ be the numbers of blocks before and after the giant, and let $T_{L,n}, T_{R,n}$ be their total lengths. Then

$$K_n = K_{L,n} + K_{R,n} + 1, \quad n - M_{1,n} = T_{L,n} + T_{R,n}.$$

Theorem 12.1 (Structural coexistence). *With $\pi_s = (1 + e^{-s})^{-1}$,*

$$\mathbb{P}(\mathcal{C}_n) \longrightarrow \pi_s, \quad \mathbb{P}(\mathcal{F}_n) \longrightarrow 1 - \pi_s. \quad (69)$$

Moreover,

$$\left(\frac{mK_n}{n}, \frac{M_{1,n}}{n} \right) \xrightarrow{d} \pi_s \delta_{(0,1)} + (1 - \pi_s) \delta_{(1,0)}. \quad (70)$$

These conclusions hold uniformly for s in a fixed compact interval in the sense of bounded continuous test functions.

Theorem 12.2 (Complete residual law on the giant event). *For each side independently, draw a count G_δ with*

$$\mathbb{P}(G_\delta = k) = \delta(1 - \delta)^k, \quad k \geq 0,$$

and then draw that many independent block lengths with law q . The joint law of the actual left and right length strings, conditional on $\mathcal{C}_n$, is within $O(1/\tau_n)$ in total variation of the product of these two string laws with $\delta = \delta_n$.

Consequently, if E_L, E_R are independent exponential variables of mean one and $G = E_L + E_R$, then, conditionally on $\mathcal{C}_n$,

$$(\delta_n K_{L,n}, \delta_n K_{R,n}, \delta_n T_{L,n}/m, \delta_n T_{R,n}/m) \xrightarrow{d} (E_L, E_R, E_L, E_R), \quad (71)$$

and

$$\left(\delta_n K_n, \frac{\delta_n}{m}(n - M_{1,n}) \right) \xrightarrow{d} (G, G), \quad G \sim \text{Gamma}(2, 1). \quad (72)$$

The density of G is $xe^{-x}\mathbf{1}_{\{x>0\}}$.

Theorem 12.3 (Stable block-count fluctuations on the other event). *Let Z be the real random variable with characteristic function*

$$\mathbb{E}e^{itZ} = \exp\{(-it)^{3/2}\}, \quad t \in \mathbb{R}, \quad (73)$$

using the principal power. Put $a_n = (\kappa n/m)^{2/3}$. Conditionally on $\mathcal{F}_n$,

$$\frac{m(K_n - n/m)}{a_n} \xrightarrow{d} -Z, \quad \frac{M_{1,n}}{n} \xrightarrow{\mathbb{P}} 0. \quad (74)$$

In fact, the convergence is joint with

$$\left(\frac{m(K_n - n/m)}{a_n}, \frac{M_{1,n}}{a_n} \right) \xrightarrow{d} (-Z, V), \quad \mathbb{P}(V \leq y) = \exp\left(-\frac{1}{2\sqrt{\pi}y^{3/2}}\right) \quad (y > 0). \quad (75)$$

The pair (Z, V) is constructed from the same Poisson point process in Section 16.3; its coordinates are dependent. In particular $K_n/n \xrightarrow{\mathbb{P}} 1/m$ on $\mathcal{F}_n$. The stable variable is centered and spectrally positive; its negative appears because a positive excess in a sum of block lengths decreases the number of blocks needed to reach n .

Two further fluctuation theorems sharpen the description of $\mathcal{C}_n$. They are proved in Section 17.

Theorem 12.4 (Residual fluctuation and second maximum). *Conditionally on $\mathcal{C}_n$, jointly,*

$$\left(\delta_n K_n, \frac{(n - M_{1,n}) - m(K_n - 1)}{(\kappa/\delta_n)^{2/3}} \right) \xrightarrow{d} (G, G^{2/3}Z), \quad (76)$$

where $G \sim \text{Gamma}(2, 1)$ and Z from (73) are independent on the right. If

$$B_n = \left(\frac{2d}{3\delta_n} \right)^{2/3}, \quad (77)$$

then, for $x > 0$,

$$\mathbb{P}(M_{2,n}/B_n \leq x \mid \mathcal{C}_n) \longrightarrow (1 + x^{-3/2})^{-2}. \quad (78)$$

More precisely, for $t \geq 0$ and $x > 0$,

$$\mathbb{E} \left[e^{-t\delta_n K_n} \mathbf{1}_{\{M_{2,n} \leq B_n x\}} \mid \mathcal{C}_n \right] \longrightarrow (1 + t + x^{-3/2})^{-2}. \quad (79)$$

These statements distinguish three scales on the giant event: n for the giant, $n/\log n$ for its deficit and the total number of remaining blocks, and $(n/\log n)^{2/3}$ for the largest remaining blocks and the centered fluctuation of their total length. On the other event the count and the largest block fluctuate on the larger scale $n^{2/3}$.

13 Analytic foundations and the uniform coefficient estimate

13.1 The singular part comes from one divisor

Separating $d = j$ in (61) gives

$$A(z) = F(z) + R(z), \quad F(z) = \frac{1}{2} \sum_{j \geq 1} \binom{2j}{j} \frac{z^j}{j^2}. \quad (80)$$

Every proper divisor of j is at most $j/2$. Bounding each binomial coefficient by $4^{j/2}$ and using at most j divisors shows that $|[z^j]R(z)| \leq 2^{j-1}$. Hence R is analytic for $|z| < 1/2$. The function F continues across the circle $|z| = 1/4$ except at $z = 1/4$, with a slit along the positive real direction.

For $t = \sqrt{1 - 4z}$, differentiation of the binomial series gives

$$\frac{dF}{dt} = -\frac{2t}{1 - t^2} \log \frac{2}{1 + t}.$$

The derivative has a convergent power series at $t = 0$. Its first odd contribution after integration is $2t^3/3$. With $w = 1 - y$ this gives

$$A(y/4) = \lambda - \mu w + \frac{2}{3}w^{3/2} + O(w^2), \quad (81)$$

$$Q(y) = 1 - mw + \kappa w^{3/2} + O(w^2). \quad (82)$$

The expansions are convergent local Puiseux expansions on a slit neighborhood, not just asymptotic statements along the real axis.

Lemma 13.1 (Tail and aperiodicity). *The critical law has finite mean m , satisfies*

$$q_j = dj^{-5/2}(1 + O(j^{-1})), \quad \mathbb{P}(X > x) \sim \frac{2d}{3}x^{-3/2}, \quad (83)$$

and has infinite variance. Also $q_1, q_3, q_4 > 0$, and $|Q(e^{it})| < 1$ for $0 < |t| \leq \pi$.

Proof. For noninteger β ,

$$[y^j](1-y)^\beta = \frac{\Gamma(j-\beta)}{\Gamma(-\beta)\Gamma(j+1)} = \frac{j^{-\beta-1}}{\Gamma(-\beta)}(1 + O(j^{-1})).$$

Apply this identity to the first two nonanalytic terms of the convergent Puiseux expansion of Q . A contour dented around $y = 1$, with its remaining part outside the unit circle, bounds the omitted terms at the corresponding higher powers. Analytic integer powers contribute no leading tail. Thus the leading coefficient is $\kappa/\Gamma(-3/2) = d$, and the next possible relative order is $1/j$. Summing the tail gives the second assertion. The moment claims follow by comparison with powers of j .

The counts $I_1 = I_3 = I_4 = 1$ give the three positive probabilities. Equality in the triangle inequality for $Q(e^{it})$ would force $e^{it} = e^{3it} = e^{4it}$. Their exponent differences have greatest common divisor one, so $t = 0$ modulo 2π . $\square$

13.2 Two terms with an absolute remainder

The next proposition is the part of the earlier report needed here. We give the proof to keep the dependency on an unrefereed analytic argument reviewable.

[write] **Added 3 October 2026, batch 85: a re-proof of Part I.** Proposition 13.2 and Corollary 13.3 are not new results. Proposition 13.2 is Theorem 4.1 of Part I, with (85) equal to (32) and (86) the first two terms of (39) (note $1/\Gamma(-3/2) = 3/(4\sqrt{\pi})$); the manuscript's "earlier report" is Part I. The normalizations agree exactly: with $r = (1 + m\tau/n)^{-1}$ and $u = r/p$ one has $u = u_n(\tau)$ of (30) (because $pm = c$), so $Z_n(r) = 4^{-n}W_n(u_n(\tau))$, $rm = a_n(\tau)$ and $\kappa/m = k = 2/(3\mu)$. Corollary 13.3 at $\tau = b_n + s$ is Corollary 6.1: $m = a_c$ and $4d/m^2 = 2/(a_c\mu\sqrt{\pi})$, so (92) is (48) divided by $a_c\sqrt{n}(\log n)^2$; its first display (90), for any $\tau \asymp \log n$ in the window, is the same two-term reading of (31). The manuscript's proof below follows Part I's Section 4 step for step ((87) is (34), (88) is (35), (89) is (38), and the zero exclusion on the circle is the same aperiodicity argument); it is printed, as delivered, so that Part II can be checked on its own, and it is not a second route.

Proposition 13.2 (Uniform logarithmic expansion; Theorem 4.1 restated). *For fixed $C > 0$, put $r = (1 + m\tau/n)^{-1}$. Uniformly for $0 \leq \tau \leq C \log n$,*

$$Z_n(r) = \frac{1}{rm} \left\{ e^{-\tau} + \frac{\kappa}{m} n^{-1/2} H(\tau) \right\} + O_C(n^{-1}), \quad (84)$$

where

$$H(\tau) = \sum_{j \geq 0} \frac{(-1)^j (j+1) \tau^j}{\Gamma(j+1/2)} = \frac{{}_1F_1(2; 1/2; -\tau)}{\sqrt{\pi}}. \quad (85)$$

As $\tau \rightarrow +\infty$,

$$H(\tau) = \frac{1}{\Gamma(-3/2)\tau^2} \left(1 + \frac{5}{\tau} + O(\tau^{-2}) \right). \quad (86)$$

Proof. Let $w = 1 - y$ and $q = \tau/n$. Equation (82) gives

$$1 - rQ(y) = rm \left(w + q - \frac{\kappa}{m} w^{3/2} + O(w^2) \right).$$

Fix θ with $\pi/2 < \theta < \pi$. On $|\arg w| \leq \theta$,

$$|w + q| \geq \cos(\theta/2)(|w| + q) \quad (q \geq 0). \quad (87)$$

Consequently the $w^{3/2}$ perturbation is uniformly smaller than half the leading term in a sufficiently small fixed sector. Lemma 13.1 excludes denominator zeros on the unit circle away from 1. Compactness and analytic continuation thicken that part of the contour a little outside the circle, uniformly for $r \rightarrow 1$. Together these observations supply a common coefficient contour: local rays $w = te^{\pm i\theta}$, a joining arc of radius $1/n$ on their right, and outer pieces on which $|y| > 1 + \epsilon$. No assertion about a whole zero-free disk beyond the circle is needed.

Expanding the resolvent on this contour gives

$$\frac{1}{1 - rQ(y)} = \frac{1}{rm(w + q)} + \frac{\kappa}{rm^2} \frac{w^{3/2}}{(w + q)^2} + E_n(w), \quad (88)$$

where

$$|E_n(w)| \leq C_1 \left(\frac{|w|^2}{(|w| + q)^2} + \frac{|w|^3}{(|w| + q)^3} \right) \leq 2C_1.$$

On each local ray the coefficient kernel is bounded by $C_2 \exp(-cn|w|)$. Its integral against E_n is $O(n^{-1})$; the joining arc has the same order of length. The outer contribution is exponentially small.

The first coefficient in (88) is exactly $(1 + \tau/n)^{-n-1}$. Its replacement by $e^{-\tau}$ costs $O(n^{-1})$ uniformly on the stated window. For the second coefficient, use $z = nw$. With a fixed-angle Hankel contour $\mathcal{H}$ around the negative real direction, indented by a unit arc to its right,

$$[y^n] \frac{w^{3/2}}{(w + q)^2} = n^{-1/2} \frac{1}{2\pi i} \int_{\mathcal{H}} e^z \frac{z^{3/2}}{(z + \tau)^2} dz + O(n^{-3/2}). \quad (89)$$

Indeed (87) bounds the integrand apart from its exponential by $C|z|^{-1/2}$. On $|z| \leq n^\eta$, $\eta < 1/2$, replacing $(1 - z/n)^{-n-1}$ by e^z has an integrable relative error $O((|z| + |z|^2)/n)$, and the rest of the rays decays exponentially. For $|\tau| < 1$, expand $(z + \tau)^{-2}$ and integrate term by term; the reciprocal-gamma Hankel identity gives (85). Continuation along $\tau \geq 0$ is valid by (87). The pole at $z = -\tau$ remains on the excluded negative direction throughout.

Finally expand $(z + \tau)^{-2}$ on the part $|z| \leq \tau/2$ of the same contour. The complementary rays have exponential decay. The first two integrals are $\tau^{-2}/\Gamma(-3/2)$ and $-2\tau^{-3}/\Gamma(-5/2)$; their ratio gives the coefficient 5. This proves (86) and the proposition. $\square$

Corollary 13.3 (The two positive contributions; Corollary 6.1 restated). *If $\tau \asymp \log n$ within a fixed logarithmic window, then*

$$Z_n(r) = \frac{e^{-\tau}}{m} (1 + o(1)) + \frac{d}{m^2} \frac{n^{-1/2}}{\tau^2} (1 + o(1)). \quad (90)$$

At $\tau = b_n + s$, set

$$D_n = \frac{4d}{m^2} \frac{n^{-1/2}}{(\log n)^2}. \quad (91)$$

Then, uniformly for bounded s ,

$$Z_n(r_n) = D_n(1 + e^{-s} + o(1)), \quad \frac{e^{-\tau_n}}{m} = D_n e^{-s}. \quad (92)$$

Proof. Use $r = 1 + O(\log n/n)$ and $d = \kappa/\Gamma(-3/2)$ in Proposition 13.2. The absolute error $O(n^{-1})$ is $o(n^{-1/2}/\log^2 n)$. Substitution of (67) proves the second statement. $\square$

14 An elementary total variation theorem for a giant block

This section is independent of the analytic form of Q . It isolates the reason that two exponential variables, and hence a Gamma law, appear.

Definition 14.1 (Geometric renewal string). For a probability mass function q on the positive integers and $0 < \delta < 1$, a geometric renewal string has law

$$\nu_\delta(x_1, \dots, x_k) = \delta(1 - \delta)^k \prod_{i=1}^k q_{x_i}, \quad k \geq 0. \quad (93)$$

The empty string is included. Formula (93) is normalized because its sum over strings of length k is $\delta(1 - \delta)^k$.

Theorem 14.2 (Giant deletion for a changing geometric weight). *Let $q_j = dj^{-1-\alpha}(1 + O(j^{-1}))$, where $d > 0$ and $\alpha > 1$, and let its mean m be finite. Give an ordered string of total size n weight $r^k \prod_i q_{x_i}$, where $r = 1 - \delta$ may depend on n . Suppose $\delta \rightarrow 0$ and $n\delta \rightarrow \infty$. Let C be the event that a block exceeds $n/2$, and let $B_n(r)$ be its unnormalized weight. Then*

$$B_n(r) = \frac{rq_n}{\delta^2} \left(1 + O\left(\frac{1}{n\delta} + \frac{1}{n}\right) \right). \quad (94)$$

The law of the two strings obtained by deleting this unique block, conditional on C , has total variation distance

$$O\left(\frac{1}{n\delta} + \frac{1}{n}\right) \quad (95)$$

from $\nu_\delta \otimes \nu_\delta$.

Proof. Choose independent left and right strings with this product law. Let T be their combined total size and J their combined number of blocks. Inserting a block of size $n - T$ between them gives an ordered string in C precisely when $T < n/2$. The location of the large block is then determined uniquely. Hence

$$B_n(r) = \frac{rq_n}{\delta^2} \mathbb{E}h_n(T), \quad h_n(T) = \frac{q_{n-T}}{q_n} \mathbf{1}_{\{T < n/2\}}. \quad (96)$$

Furthermore, the conditional residual law has density $h_n(T)/\mathbb{E}h_n(T)$ with respect to the product string law. These are exact identities, before any limiting argument.

For $0 \leq t < n/2$, the assumed tail gives, uniformly in t ,

$$\frac{q_{n-t}}{q_n} = (1 - t/n)^{-1-\alpha}(1 + O(n^{-1})), \quad \left| \frac{q_{n-t}}{q_n} - 1 \right| \leq C \left(\frac{t}{n} + \frac{1}{n} \right).$$

Each string has expected size mr/δ , so $\mathbb{E}T = 2mr/\delta$. On $T \geq n/2$, $|h_n - 1| = 1$ and Markov's inequality applies. Therefore

$$\mathbb{E}|h_n(T) - 1| \leq C' \left(\frac{\mathbb{E}T}{n} + \frac{1}{n} \right) = O\left(\frac{1}{n\delta} + \frac{1}{n}\right). \quad (97)$$

This proves (94). Finally,

$$d_{\text{TV}} \left(\frac{h_n}{\mathbb{E}h_n} \nu_\delta^{\otimes 2}, \nu_\delta^{\otimes 2} \right) = \frac{1}{2} \mathbb{E} \left| \frac{h_n}{\mathbb{E}h_n} - 1 \right| \leq \frac{\mathbb{E}|h_n - 1|}{\mathbb{E}h_n},$$

which proves (95). $\square$

Only regular variation, rather than the displayed $O(j^{-1})$ error, is needed for an $o(1)$ total variation statement: on $t < n/2$ the ratio is bounded, and $T/n \xrightarrow{\mathbb{P}} 0$. The stronger local tail is used to give the explicit rate.

14.1 Gamma laws and the location of the giant

Let N_δ be the count in one geometric string, and let T_δ be its total size. Then

$$\delta N_\delta \xrightarrow{d} E, \quad E \sim \text{Exp}(1). \quad (98)$$

Also

$$\delta(T_\delta - mN_\delta) \xrightarrow{\mathbb{P}} 0. \quad (99)$$

For completeness, choose $A > 0$. The probability $N_\delta > A/\delta$ can be made uniformly small by increasing A . By the ordinary strong law, $\max_{k \leq L} |\sum_{i \leq k} (X_i - m)|/L \rightarrow 0$ almost surely as $L \rightarrow \infty$. Apply this with $L = \lceil A/\delta \rceil$ to obtain (99). The argument uses only a finite first moment.

Theorem 14.2, with $\alpha = 3/2$ and $n\delta_n \asymp \tau_n$, now proves Theorem 12.2. It proves convergence of the entire strings, so every bounded statistic of them can be transferred; the statistic may itself depend on n .

Corollary 14.3 (A uniform split around the giant). *Conditional on $\mathcal{C}_n$, jointly with $\delta_n K_n \xrightarrow{d} G$,*

$$\frac{K_{L,n}}{K_n - 1} \xrightarrow{d} U, \quad \frac{T_{L,n}}{n - M_{1,n}} \xrightarrow{d} U, \quad (100)$$

where U is uniform on $(0, 1)$ and independent of G . Ratios with zero denominator can be assigned any value.

Proof. The four-coordinate limit (71) reduces both ratios to $E_L/(E_L + E_R)$. The change of variables $(E_L, E_R) = (GU, G(1 - U))$ has Jacobian G and density Ge^{-G} on $G > 0$, $0 < U < 1$. Thus U is uniform and independent of G . The limiting denominator is positive almost surely. $\square$

15 Identification of the phase probabilities

For the tournament law, the exact marked-block identity in (96) is equivalently

$$B_n(r) = r \sum_{t < n/2} q_{n-t}[y^t] \frac{1}{(1 - rQ(y))^2}. \quad (101)$$

It sums over the unique block exceeding half the total size. It does not confuse a count of marked blocks with the probability of an event containing several possible marks.

Theorem 14.2 and (68) give

$$B_n(r_n) \sim \frac{r_n q_n}{\delta_n^2} \sim \frac{d}{m^2} \frac{n^{-1/2}}{\tau_n^2} \sim D_n. \quad (102)$$

Divide this by (92). The result is $\mathbb{P}(\mathcal{C}_n) \rightarrow \pi_s$, proving (69). On $\mathcal{C}_n$, Theorem 12.2 implies $K_n/n \xrightarrow{\mathbb{P}} 0$ and $M_{1,n}/n \xrightarrow{\mathbb{P}} 1$ because $n\delta_n \rightarrow \infty$.

15.1 A short independent check from a probability generating function

The limiting block-count mixture can also be read from a controlled perturbation of the parameter. For fixed $t \geq 0$,

$$\mathbb{E}e^{-tK_n/n} = \frac{Z_n(r_n e^{-t/n})}{Z_n(r_n)}. \quad (103)$$

Write $r_n e^{-t/n} = (1 + m\tau'_n/n)^{-1}$. Exactly,

$$\tau'_n = \frac{n}{m} \left[(1 + m\tau_n/n) e^{t/n} - 1 \right] = \tau_n + t/m + o(1).$$

Uniformity in Corollary 13.3 yields

$$\mathbb{E}e^{-tK_n/n} \rightarrow \frac{1 + e^{-s-t/m}}{1 + e^{-s}}. \quad (104)$$

This is the Laplace transform of atoms at 0 and $1/m$. Since $0 \leq K_n/n \leq 1$, it identifies the weak limit and also all its moments. The argument needs the expansion for nearby parameters; an uncontrolled decomposition at one value would not suffice.

The giant event already accounts for the whole limiting mass at $K_n/n = 0$. Subtracting its bounded-test-function limits from (104) proves $K_n/n \xrightarrow{\mathbb{P}} 1/m$ on $\mathcal{F}_n$.

15.2 Exclusion of intermediate macroscopic blocks

Fix $\epsilon > 0$, and choose $\eta > 0$ so small that $m\eta < \epsilon/4$. For

$$k \geq n/m - \eta n,$$

a specified block of size at least ϵn leaves at most $(1 - \epsilon)n$ for the other $k - 1$ blocks. This is a fixed proportion below their mean for large n . Since $X \geq 1$, its negative exponential moment exists. The finite mean expansion

$$\log \mathbb{E}e^{-hX} = -mh + o(h) \quad (h \downarrow 0)$$

shows, by choosing one sufficiently small fixed $h > 0$, that

$$\mathbb{P}(X_1 + \cdots + X_{k-1} \leq (1 - \epsilon)n) \leq e^{-c_\epsilon n} \quad (105)$$

uniformly over these $k \leq n$. Marking the exceptional block, summing over its size and location, and over k , gives an unnormalized weight at most $n^2 e^{-c_\epsilon n}$, since $r_n^k \leq 1$ and $\sum_j q_j = 1$. Division by the polynomially small $Z_n(r_n)$ still gives a vanishing probability.

On $\mathcal{F}_n$, the complementary range of k has probability tending to zero by the preceding count limit. Thus $M_{1,n}/n \xrightarrow{\mathbb{P}} 0$ on $\mathcal{F}_n$. This completes the proof of Theorem 12.1. The same estimates, with parameters on a compact s -interval, give the asserted uniformity.

Corollary 15.1 (Macroscopic moments). *Let J_s be Bernoulli with $\mathbb{P}(J_s = 1) = 1 - \pi_s$. Then $mK_n/n \xrightarrow{d} J_s$ and $M_{1,n}/n \xrightarrow{d} 1 - J_s$, jointly. Consequently,*

$$\begin{aligned} \frac{\mathbb{E}K_n}{n} &\longrightarrow \frac{1 - \pi_s}{m}, & \frac{\text{Var } K_n}{n^2} &\longrightarrow \frac{\pi_s(1 - \pi_s)}{m^2}, \\ \frac{\text{Cov}(K_n, M_{1,n})}{n^2} &\longrightarrow -\frac{\pi_s(1 - \pi_s)}{m}. \end{aligned} \quad (106)$$

Proof. The normalized variables are bounded, so weak convergence implies convergence of the indicated moments. This does not assert moment convergence for the unbounded stable or Gamma rescalings. $\square$

16 A direct stable local limit theorem and the fluid law

16.1 Local limit by Fourier inversion

Lemma 16.1 (Lattice 3/2-stable local limit). *Let $S_k^X = X_1 + \dots + X_k$ and $a(k) = (\kappa k)^{2/3}$. The variable Z in (73) has a continuous density g , and*

$$\sup_{\ell \in \mathbb{Z}} \left| a(k) \mathbb{P}(S_k^X = \ell) - g\left(\frac{\ell - mk}{a(k)}\right) \right| \longrightarrow 0. \quad (107)$$

Proof. Put $\phi(t) = Q(e^{it})$. The expansion (82) implies

$$\log \phi(t) = imt + \kappa(-it)^{3/2} + O(t^2). \quad (108)$$

Indeed $1 - e^{it} = -it + O(t^2)$, and its principal 3/2 power has the indicated leading term. For real t ,

$$(-it)^{3/2} = -\frac{|t|^{3/2}}{\sqrt{2}}(1 + i \operatorname{sgn} t).$$

It follows that $|\phi(t)| \leq \exp(-c|t|^{3/2})$ near zero. Away from zero Lemma 13.1 gives a strict uniform bound below one.

The characteristic functions of $(S_k^X - mk)/a(k)$ converge pointwise to $\exp\{(-it)^{3/2}\}$, by (108); their limit is continuous at zero and hence is a characteristic function. It is integrable, so its inverse Fourier transform is a continuous density g of total integral one.

Lattice inversion, followed by $v = a(k)t$, writes $a(k)\mathbb{P}(S_k^X = \ell)$ as the integral over $[-\pi a(k), \pi a(k)]$ of

$$\frac{1}{2\pi} e^{-iv(\ell - mk)/a(k)} \left[e^{-imv/a(k)} \phi(v/a(k)) \right]^k.$$

The bracketed kernel converges in $L^1(dv)$ to $\exp\{(-iv)^{3/2}\}$. On the shrinking small-angle region it is bounded by $\exp(-c'|v|^{3/2})$; on the remaining angles the integral is at most $O(a(k)\gamma^k)$ for some $\gamma < 1$. Dominated convergence proves the L^1 assertion. The oscillatory factor has modulus one for every ℓ , so it also proves the uniform bound (107). $\square$

16.2 Central windows and exhaustion of the remaining mass

Let $a_n = a(n/m) = (\kappa n/m)^{2/3}$, interpreting the formula at a real argument, and define

$$x_{n,k} = \frac{m(k - n/m)}{a_n}.$$

For $x_{n,k}$ in any fixed bounded interval, $a(k)/a_n \rightarrow 1$ uniformly and (107) gives

$$\mathbb{P}(S_k^X = n) = a_n^{-1}(g(-x_{n,k}) + o(1)). \quad (109)$$

Also, uniformly on this window,

$$\frac{r_n^k}{e^{-\tau_n}} \rightarrow 1. \quad (110)$$

To check the latter, use $\log r_n = -m\tau_n/n + O(\tau_n^2/n^2)$ and $k = n/m + O(n^{2/3})$. The remaining exponent is $O(\tau_n n^{-1/3} + \tau_n^2/n) = o(1)$.

For fixed $A < B$, a Riemann sum in (109) yields

$$\sum_{A \leq x_{n,k} \leq B} r_n^k \mathbb{P}(S_k^X = n) = \frac{e^{-\tau_n}}{m} \left(\int_A^B g(-x) dx + o(1) \right). \quad (111)$$

The mesh of $x_{n,k}$ is m/a_n , explaining the factor $1/m$.

The contribution in this window having a block greater than $n/2$ is exponentially small by (105). Hence (111) holds with the event $\mathcal{F}_n$ imposed. On the other hand, (92) and (102) imply

$$Z_n(r_n) - B_n(r_n) \sim \frac{e^{-\tau_n}}{m}. \quad (112)$$

This subtraction is legitimate because s is fixed, so both limiting contributions have positive finite relative weights.

Divide (111) by (112). Letting $A \rightarrow -\infty$ and $B \rightarrow \infty$ exhausts mass one because g is a probability density. This proves tightness of the conditional rescaled count as well as convergence on bounded intervals. The limit has density $g(-x)$, which is precisely the law of $-Z$. Together with Section 15, this proves Theorem 12.3.

The exhaustion argument is useful: a local limit theorem alone does not control all possible intermediate ranges. The separately proved giant weight and total coefficient show that no additional range has unaccounted mass at coexistence.

16.3 Largest fluid block and its dependence on the count

Let $\mathcal{P}$ be a Poisson point process on $(0, \infty)$ with intensity

$$\nu(dv) = \frac{v^{-5/2}}{\Gamma(-3/2)} dv.$$

Let V be its largest point and define the compensated sum

$$Z = \lim_{\epsilon \downarrow 0} \left(\sum_{\substack{v \in \mathcal{P} \\ v > \epsilon}} v - \int_{\epsilon}^{\infty} v \nu(dv) \right). \quad (113)$$

The compensated contribution of points below one converges in L^2 because $\int_0^1 v^2 \nu(dv) < \infty$. Above one there are finitely many points almost surely, and their sum has finite mean. The exponential formula gives

$$\mathbb{E} e^{itZ} = \exp \left\{ \int_0^\infty (e^{itv} - 1 - itv) \nu(dv) \right\} = \exp \{ (-it)^{3/2} \}.$$

The last integral follows by two integrations by parts, or the gamma integral with a positive real damping parameter followed by a limit. Thus this is the same stable law as before. Also

$$\mathbb{P}(V \leq y) = \exp \{ -\nu((y, \infty)) \} = \exp \left(-\frac{1}{2\sqrt{\pi} y^{3/2}} \right).$$

We prove the joint limit (75), including its dependence. For $y > 0$, define the defective density g_y by

$$\int_{\mathbb{R}} e^{itx} g_y(x) dx = \Psi_y(t) = \exp \left\{ (-it)^{3/2} - \int_y^\infty e^{itv} \nu(dv) \right\}. \quad (114)$$

The Poisson formula identifies the right side as $\mathbb{E}[e^{itZ} \mathbf{1}_{\{V \leq y\}}]$. It is integrable: its modulus is bounded by $\exp \{ \nu((y, \infty)) - |t|^{3/2}/\sqrt{2} \}$. Consequently g_y is continuous, nonnegative, and has total mass $\mathbb{P}(V \leq y)$.

Truncate the one-block characteristic function at $\lfloor ya_n \rfloor$:

$$\phi_{n,y}(t) = \sum_{j \leq ya_n} q_j e^{itj}.$$

For $k = n/m + O(a_n)$, regular variation and a Riemann sum imply

$$k \sum_{j > ya_n} q_j e^{itj/a_n} \longrightarrow \frac{1}{\Gamma(-3/2)} \int_y^\infty e^{itv} v^{-5/2} dv.$$

Combining this with (108) shows

$$e^{-itm k/a_n} \phi_{n,y}(t/a_n)^k \longrightarrow \Psi_y(t).$$

Convergence is uniform on bounded count windows. The Fourier kernels again converge in L^1 . Near zero, removal of the tail changes the characteristic function by at most C_y/n , and hence

$$|\phi_{n,y}(t/a_n)|^k \leq C'_y \exp(-c'|t|^{3/2})$$

for small fixed $|t|/a_n$. One may obtain this bound by applying $|\phi_{n,y}| \leq |\phi| + C_y/n$ to the exponential bound used in Lemma 16.1. Away from zero, the strict aperiodicity bound still applies for large n .

Fourier inversion therefore gives, uniformly on bounded count windows,

$$a_n \mathbb{P}(S_k^X = n, \max_{i \leq k} X_i \leq ya_n) = g_y(-x_{n,k}) + o(1).$$

After multiplying by r_n^k and taking a Riemann sum, its weight on a fixed interval I of count coordinates is

$$\frac{e^{-\tau_n}}{m} \left(\int_I g_y(-x) dx + o(1) \right).$$

For fixed y this event lies in $\mathcal{F}_n$ for large n , because $a_n = o(n)$. Divide by (112). The already proved conditional tightness of the count permits I to increase to all of $\mathbb{R}$. This yields the stated Fréchet

marginal; bounded rectangles identify the joint law. The count and maximum marginals give joint tightness, completing the proof.

The compensation term in (114) is necessary. Simply discarding the points above y would change the deterministic centering as well as the probability that no such point occurs. The largest fluid block and the stable count fluctuation are thus two observables of one limiting process.

The argument also identifies every large fluid block jointly. For nonnegative continuous f of compact support in $(0, \infty)$, replace the truncated characteristic function by $\sum_j q_j \exp(itj/a_n - f(j/a_n))$. The perturbation is $O(n^{-1})$ and has exactly the same Fourier domination. Its limiting exponent is

$$(-it)^{3/2} - \int_0^\infty (1 - e^{-f(v)}) e^{itv} \nu(dv).$$

The same local inversion, count-window summation, and tightness argument prove joint convergence of the rescaled count and the point measure of block sizes to $(-Z, \mathcal{P})$, with vague convergence on $(0, \infty)$. Here the mark $\exp(-\sum f)$ is at most one, so inserting $\mathcal{F}_n$ changes the central-window expectations by at most the exponentially small giant contamination in (105). In particular, any fixed number of largest fluid blocks converges to the corresponding ranked points. This is a statement about sizes; a functional limit keeping their positions in the ordered sequence is a separate question.

17 Finer laws inside the giant phase

17.1 Stable fluctuations at a Gamma random time

Let N_δ be the count in one independent geometric string. For $t \geq 0$ and $\theta \in \mathbb{R}$, its exact transform is

$$\mathbb{E} \exp \left\{ -t\delta N_\delta + i\theta \frac{T_\delta - mN_\delta}{(\kappa/\delta)^{2/3}} \right\} = \frac{\delta}{1 - re^{-t\delta} \mathbb{E} \exp \{ i\theta(X - m)(\delta/\kappa)^{2/3} \}}. \quad (115)$$

Equation (108) says that the last expectation is $1 + \delta(-i\theta)^{3/2} + o(\delta)$. The denominator divided by δ tends to $1 + t - (-i\theta)^{3/2}$. Squaring for two independent strings gives

$$(1 + t - (-i\theta)^{3/2})^{-2}. \quad (116)$$

This is the joint transform of $(G, G^{2/3}Z)$: conditional on G , the second coordinate has characteristic function $\exp\{G(-i\theta)^{3/2}\}$. Total variation transfer and $\delta_n \rightarrow 0$ permit replacing the residual count $K_n - 1$ by K_n in the first coordinate. This proves (76).

The variables in (72) should therefore not be read as independent. Their leading limits are the same random variable. Equation (76) measures the smaller fluctuation around that common leading value.

17.2 Second-largest block and joint maximum law

Let $F(x) = \mathbb{P}(X \leq x)$ and $B(\delta) = ((2d/3)/\delta)^{2/3}$. By (83),

$$1 - F(B(\delta)x) = \delta x^{-3/2}(1 + o(1)) \quad (x > 0). \quad (117)$$

If V_δ is the largest size in one geometric string, with maximum zero for the empty string, summing the geometric series gives

$$\mathbb{E}[e^{-t\delta N_\delta} \mathbf{1}_{\{V_\delta \leq B(\delta)x\}}] = \frac{\delta}{1 - re^{-t\delta} F(B(\delta)x)} \longrightarrow \frac{1}{1 + t + x^{-3/2}}. \quad (118)$$

The maximum of the two residual strings is the second-largest block of the original object on $\mathcal{C}_n$. Squaring (118) and applying Theorem 12.2 proves (78)–(79).

There is a useful probabilistic reading: conditional on $G = g$, the limiting rescaled second maximum has distribution $\exp(-gx^{-3/2})$. Mixing this Fréchet distribution over $G \sim \text{Gamma}(2, 1)$ produces $(1 + x^{-3/2})^{-2}$. Thus the second maximum and the total residual count are dependent.

17.3 All large residual blocks as a point process

The same computation describes more than the largest residual block. For a nonnegative continuous function f with compact support in $(0, \infty)$, set

$$J_f = \int_0^\infty (1 - e^{-f(x)}) \frac{3}{2} x^{-5/2} dx.$$

Regular variation gives

$$\delta^{-1} \mathbb{E}[1 - e^{-f(X/B(\delta))}] \longrightarrow J_f.$$

Repeating the geometric sum with this factor instead of F yields

$$\mathbb{E} \left[\exp \left\{ -t\delta_n K_n - \sum_{j \geq 2} f(M_{j,n}/B_n) \right\} \mid \mathcal{C}_n \right] \longrightarrow (1 + t + J_f)^{-2}. \quad (119)$$

This specifies the Laplace functionals of the limiting point process. Given G , it is a Poisson point process on $(0, \infty)$ with intensity $G(3/2)x^{-5/2}dx$. This statement concerns points bounded away from zero. It does not assert that uncentered sums of all limiting points are finite; the intensity has a divergent first moment at zero, and centered stable fluctuations require compensation.

17.4 All fixed inverse-logarithmic corrections

Here $(a)_j = a(a+1)\cdots(a+j-1)$ and $(a)_0 = 1$. The expansion of the heavy-tail kernel has a direct interpretation in terms of the gamma residual geometry; the coefficients are not merely formal contour artifacts.

Proposition 17.1 (All fixed orders for the giant weight). *Fix $C_0 > 0$ and an integer $M \geq 0$. Let $r = (1 + m\tau/n)^{-1}$ and $\delta = 1 - r$, with $1 \leq \tau \leq C_0 \log n$. Then, uniformly in this range,*

$$B_n(r) = \frac{rq_n}{\delta^2} \left\{ \sum_{j=0}^M (j+1)(5/2)_j \tau^{-j} + O_{M,C_0}(\tau^{-M-1}) \right\}.$$

In particular the expansion begins with $1 + 5/\tau + 105/(4\tau^2)$, and its prefactor can equivalently be written as $(d/(rm^2))n^{-1/2}\tau^{-2}(1 + O(n^{-1}))$.

Proof. First, the tail estimate gives an elementary first absolute moment bound for centered sums:

$$\mathbb{E}|S_k^X - mk| \leq Ck^{2/3} \quad (k \geq 1).$$

Indeed, take $A = k^{2/3}$ and split each summand into its part at most A and its remaining part. The assumed mass-function bound gives

$$\mathbb{E}[X\mathbf{1}_{X>A}] \leq CA^{-1/2}, \quad \mathbb{E}[X^2\mathbf{1}_{X\leq A}] \leq CA^{1/2}.$$

The centered truncated sum has first absolute moment at most its standard deviation, $\sqrt{CkA^{1/2}}$, and the centered tail sum has first absolute moment at most $2CkA^{-1/2}$. Both are $O(k^{2/3})$.

Under the reference law write $N = N_L + N_R$, so T has law S_N^X . Concavity and independence imply

$$\mathbb{E}_r|T - mN| \leq C\mathbb{E}_r N^{2/3} \leq C\delta^{-2/3}.$$

Let $f(t)$ equal $(1-t)^{-5/2}$ for $0 \leq t \leq 1/2$ and equal $2^{5/2}$ for $t \geq 1/2$. This is bounded and Lipschitz. Its use also controls the discontinuous cutoff in h_n :

$$\mathbb{E}_r|h_n(T) - f(mN/n)| = O\left(n^{-1} + \frac{\delta^{-2/3}}{n} + \mathbb{P}_r(T \geq n/2)\right).$$

Moreover

$$\mathbb{P}_r(T \geq n/2) \leq \mathbb{P}_r(N \geq n/(4m)) + \frac{4\mathbb{E}_r|T - mN|}{n}.$$

For the negative-binomial variable N , the exact tail at integer $L \geq 0$ is

$$\mathbb{P}_r(N \geq L) = r^L(1 + \delta L).$$

Because $n\delta = mr\tau$ and $r \rightarrow 1$ uniformly in the stated window, this proves

$$\mathbb{E}_r|h_n(T) - f(mN/n)| = O\left(n^{-1} + n^{-1/3}\tau^{-2/3} + (1 + \tau)e^{-c\tau}\right)$$

with a fixed $c > 0$.

For every fixed M , the global Taylor-remainder bound

$$\left|f(t) - \sum_{j=0}^M \frac{(5/2)_j}{j!} t^j\right| \leq C_M t^{M+1} \quad (t \geq 0)$$

holds. On $[0, 1/2]$ it is the usual Taylor estimate; on $[1/2, \infty)$ it follows because f is bounded and the displayed polynomial has degree M . The negative-binomial factorial moments are exact:

$$\mathbb{E}_r N^j = (j+1)! \left(\frac{r}{\delta}\right)^j.$$

Converting ordinary moments to factorial moments and using $mr/(n\delta) = 1/\tau$ gives

$$\mathbb{E}_r f(mN/n) = \sum_{j=0}^M (j+1)(5/2)_j \tau^{-j} + O_M(\tau^{-M-1} + n^{-1}).$$

Finally, for $1 \leq \tau \leq C_0 \log n$, each of n^{-1} , $n^{-1/3}\tau^{-2/3}$, and $(1 + \tau)e^{-c\tau}$ is $O_{M,C_0}(\tau^{-M-1})$. Insert the resulting expansion of $\mathbb{E}_r h_n(T)$ into the exact formula $B_n(r) = r q_n \delta^{-2} \mathbb{E}_r h_n(T)$. The equivalent prefactor follows from $r q_n / \delta^2 = (d/(rm^2)) n^{-1/2} \tau^{-2} (1 + O(n^{-1}))$. $\square$

Corollary 17.2 (Explicit residual approximations to every fixed order). *Let R_r denote the product of the two geometric-string reference laws in Theorem 12.2, and write $N = N_L + N_R$. For a fixed integer $M \geq 0$, define*

$$P_M(x) = \sum_{j=0}^M \frac{(5/2)_j}{j!} x^j, \quad \frac{dR_r^{(M)}}{dR_r} = \frac{P_M(mN/n)}{\mathbb{E}_r P_M(mN/n)}.$$

This is a probability measure because the polynomial is positive on $[0, \infty)$ and N has moments of every order. If $\tau \rightarrow \infty$ and $\tau \leq C_0 \log n$, then the actual joint residual-string law $\mathbf{L}_{n,r}$, conditional on $\mathcal{C}_n$, satisfies

$$d_{\text{TV}}(\mathbf{L}_{n,r}, \mathbf{R}_r^{(M)}) = O_{M,C_0}(\tau^{-M-1}).$$

In particular, in the coexistence window this gives an explicit $O((\log n)^{-M-1})$ approximation for every fixed M . Conditionally on their two counts, the block lengths under $\mathbf{R}_r^{(M)}$ still have the independent law q .

Proof. The absolute-value estimates in the preceding proof and its global Taylor bound give

$$\mathbb{E}_r |h_n(T) - P_M(mN/n)| = O_{M,C_0}(\tau^{-M-1}).$$

Indeed the Taylor error has expectation at most $C_M(m/n)^{M+1} \mathbb{E}_r N^{M+1} = O_M(\tau^{-M-1})$. The exact residual density is $h_n(T)/\mathbb{E}_r h_n(T)$, and $\mathbb{E}_r h_n(T) = 1 + O(\tau^{-1})$ is bounded away from zero. For nonnegative integrable functions h, P ,

$$\frac{1}{2} \mathbb{E} \left| \frac{h}{\mathbb{E}h} - \frac{P}{\mathbb{E}P} \right| \leq \frac{\mathbb{E}|h - P|}{\mathbb{E}h}.$$

Applying this inequality proves the assertion. $\square$

18 Centering, finite-size effects, and a reusable principle

18.1 An implicit center that keeps the logarithmic corrections

The explicit b_n is convenient for a theorem, but the $2 \log \log n$ correction can be large at accessible sizes. A useful alternative is the large solution $\hat{b}_n > 2$ of

$$\hat{b}_n^2 e^{-\hat{b}_n} = \frac{d}{m\sqrt{n}}. \quad (120)$$

For sufficiently large n it is unique on $(2, \infty)$ because $x^2 e^{-x}$ is strictly decreasing there. In terms of the lower real Lambert branch,

$$\hat{b}_n = -2W_{-1} \left(-\frac{1}{2} \sqrt{d/m} n^{-1/4} \right). \quad (121)$$

This is the exact balance of the leading fluid and giant terms in (90), not the exact finite- n median of the giant event.

[write] **Added 3 October 2026, batch 85.** The inversion (121) is an instance of `p0:thm:lambert-core` of the repository's transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, and no novelty is claimed for it. Taking logarithms in (120) gives $x - 2 \log x = \Lambda$ with $\Lambda = \frac{1}{2} \log n + \log(m/d)$ (the equation used in the proof of Proposition 18.1); this is the theorem's equation $aX + b \log X = \Lambda$ with $a = 1$, $b = -2$, and its branch rule for $b < 0$ gives two positive solutions once $\Lambda > 2(1 - \log 2)$, the larger one, $X > |b|/a = 2$, on the branch W_{-1} at $z = (a/b)e^{\Lambda/b} = -\frac{1}{2} \sqrt{d/m} n^{-1/4}$. The order-zero inverse (51) of Part I is the same theorem with $a = \log 4$, $b = -5/2$ (the batch-77 note at the end of Section 7).

Proposition 18.1 (Refined center and error scale). *Put $c_0 = \log(m/(4d))$ and $L = \log n$. Then*

$$\hat{b}_n = b_n + \frac{8 \log L + 4c_0}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (122)$$

All the main theorems remain valid with $\tau_n = \hat{b}_n + s$. For this choice, uniformly for bounded s ,

$$\mathbb{P}(\mathcal{C}_n) = \frac{1}{1 + e^{-s}} + O\left(\frac{1}{\log n}\right). \quad (123)$$

With the elementary center $b_n + s$, the same argument gives the weaker bound $O(\log \log n / \log n)$.

Proof. Taking logarithms of (120) gives $x - 2 \log x = L/2 + \log(m/d)$. Substitute the right side of (122); expansion of $\log x$ leaves a residual $O((\log L)^2/L^2)$. The derivative $1 - 2/x$ is bounded away from zero for large x , proving the expansion by the mean value theorem. In particular $\hat{b}_n - b_n \rightarrow 0$, which preserves all the limiting distributions.

For the rate, write $D = dn^{-1/2}/(m^2 \hat{b}_n^2)$. Equation (120) makes $e^{-(\hat{b}_n + s)}/m = De^{-s}$ exactly. Theorem 14.2 gives $B_n = D(1 + O(1/\log n))$, uniformly for bounded s . Proposition 13.2 and (86) give $Z_n = D(1 + e^{-s} + O(1/\log n))$; the relative contour error $O(n^{-1/2} \log^2 n)$ is smaller. Dividing proves (123). The last assertion follows from (122) and smoothness of the logistic function. $\square$

The probability transition occurs over $O(1)$ changes of s , or $O(1/n)$ changes of u around its moving center. The center itself lies below u_c by order $\log n/n$. Confusing this displacement with the narrower transition width obscures the two scales.

18.2 What is specific to tournaments

Theorem 14.2 applies to any positive integer law with a finite mean and a regular power tail. The tournament identities supply the particular exponent $5/2$, the constants, and the uniform coefficient control. The stable exponent $3/2$ comes from the block tail, while Gamma shape two comes from having a string on each side of a unique giant.

For a law $q_j \sim dj^{-1-\alpha}$ with $1 < \alpha < 2$, a two-term renewal estimate of the corresponding form would balance

$$\frac{e^{-\tau}}{m} \quad \text{and} \quad \frac{d}{m^2} n^{1-\alpha} \tau^{-2}.$$

It would have center

$$(\alpha - 1) \log n + 2 \log \log n + \log \frac{m(\alpha - 1)^2}{d}.$$

The giant-deletion theorem would then give the same Gamma law, and the appropriate stable local limit would give count fluctuations $n^{1/\alpha}$ in the linear-block regime. This is a deduction conditional on those two analytic inputs, not a blanket theorem under regular variation alone. It states the reusable architecture without silently assuming a uniform renewal expansion for every law.

19 Computation and reproducibility

The computations accompanying this manuscript have two separate roles: exact checks of the finite combinatorial identities and numerical diagnostics of convergence. The proofs above do not rely on a numerical extrapolation.

19.1 Exact arithmetic checks

The exact recurrences are

$$\begin{aligned} nS_n &= \sum_{j=1}^n N_j S_{n-j}, \\ I_n &= S_n - \sum_{j=1}^{n-1} I_j S_{n-j}, \\ W_0(u) &= 1, \quad W_n(u) = u \sum_{j=1}^n I_j W_{n-j}(u). \end{aligned} \tag{124}$$

For comparison with direct enumeration, the full block-count polynomial is obtained by retaining u symbolically. The giant-event polynomial has the independent marked-block expression

$$\sum_{j > n/2} u I_j [z^{n-j}] (1 - uI(z))^{-2}. \tag{125}$$

The verifier enumerates the integer tuples in (59), cuts each at its equality indices, and compares both polynomials coefficient by coefficient. This checks the count, the block statistic, and the strict inequality defining the giant event.

The completed run examined 125,476 nondecreasing candidate sequences over all lengths through $n = 10$. Every exact comparison passed. At $n = 10$, the block-count polynomial is

$$573u + 433u^2 + 246u^3 + 130u^4 + 55u^5 + 33u^6 + 7u^7 + 8u^8 + u^{10},$$

and the giant-event polynomial is

$$573u + 424u^2 + 225u^3 + 84u^4 + 35u^5.$$

Their values at $u = 1$ are 1486 and 1341. Exact counts through $n = 200$ also supply independent checks of the normalized numerical recurrences.

19.2 Constants and floating diagnostics

If $R_j = [z^j]R(z)$ in (80), then

$$\lambda = \frac{\pi^2}{12} - \log^2 2 + \sum_{j \geq 1} \frac{R_j}{4^j}, \quad \mu = \log 2 + \sum_{j \geq 1} \frac{j R_j}{4^j}. \tag{126}$$

The omitted tails after M are bounded respectively by 2^{-M-1} and $(M+2)2^{-M-1}$. These are analytic truncation bounds; ordinary floating evaluations of the finite sums are still diagnostics unless their rounding is separately enclosed.

Numerically the constants begin

$$\begin{aligned} \lambda &= 0.3302375439859293 \dots, & \mu &= 0.6739526018515949 \dots, \\ u_c &= 3.5555932000209863 \dots, & m &= 1.7223486864283872 \dots \end{aligned}$$

The numerical code works with 4^{-n} -normalized coefficients to avoid exponential overflow. It evaluates $Z_n(r)$ by its positive renewal recurrence and evaluates (101) by a convolution. It compares the

resulting giant probabilities with the limiting logistic curve around $\hat{b}_n$. The default run computes normalized coefficients through $n = 8000$. In the recorded environment the long-double type has 64 significant binary digits. Through $n = 200$, its relative discrepancies from the exact counts are at most $1.1 \cdot 10^{-19}$ for S_n and $1.7 \cdot 10^{-19}$ for I_n . These comparisons are useful numerical checks, not interval error bounds for every subsequent calculation.

The measured center probabilities show why the distinction matters:

n	$\hat{b}_n$	$\mathbb{P}(\mathcal{C}_n)$ at $s = 0$
512	8.1978140113	0.5241164884
2000	9.0845092174	0.5341770504
8000	9.9620885205	0.5498246417

These values do not converge monotonically toward the limiting $1/2$. At $n = 8000$, the giant weight is about 1.73615 times its leading approximation, and the total weight is about 1.57882 times the sum of the two leading approximations. Terms such as $5/\tau$ and $105/(4\tau^2)$ in Proposition 17.1 are still substantial when τ is only about ten. No theorem above claims a small finite- n relative error from the leading terms alone.

For Figure 2, FFT polynomial powers were checked against the positive recurrence: the total-mass relative discrepancy was below $1.4 \cdot 10^{-15}$, and the giant-mass discrepancy below $1.0 \cdot 10^{-17}$. The computed conditional means are approximately 148.45 blocks on $\mathcal{C}_n$ and 468.27 on $\mathcal{F}_n$, whereas the leading scales give 134.45 and 580.60. This quantifies the visible finite-size correction rather than treating the plots as close asymptotic agreement.

The package contains the source, compiled article, verification programs, recorded outputs, and figures. It is independent of a ProveIt checkout. No assertion is made that numerical agreement establishes a total variation estimate or a distributional limit.

[write] **Added 3 October 2026, batch 85.** The preceding paragraph describes the delivered package. In this report its files carry the prefix `02-coex-`: the program `code/verify_and_plot.py` is `code/02-coex-verify_and_plot.py`, the build script is `code/02-coex-build.sh`, the code README is `02-coex-code-README.md` at the report root, the requirements and `provenance.json` are in `data/`, and every `data/` and `figures/` file keeps its name after the prefix. The delivered source, PDF, README and checksum list are not shipped; they survive in the archive of arrival commit `317c1ce2e`. Run as delivered, the program writes into `data/` and `figures/` beside `code/` under the unprefix names, so the report's `README.md` runs it on a scratch copy with an explicit `-output-root`. Such a run at intake (Windows, Python 3.12, the pinned packages) passed every exact assertion with identical exact output; its floating diagnostics agreed with the recorded ones to a relative $7.2 \cdot 10^{-13}$ (coexistence table), $5 \cdot 10^{-15}$ (normalized counts and renewal masses) and an absolute $6.5 \cdot 10^{-14}$ (block-count distribution), but not digit for digit, because on that platform `numpy.longdouble` is ordinary double precision (52 stored mantissa bits against the recorded 63), the platform dependence that the code README discloses. The figure showing the theoretical phase profiles, which the manuscript ships but does not display, is printed as Figure 3.

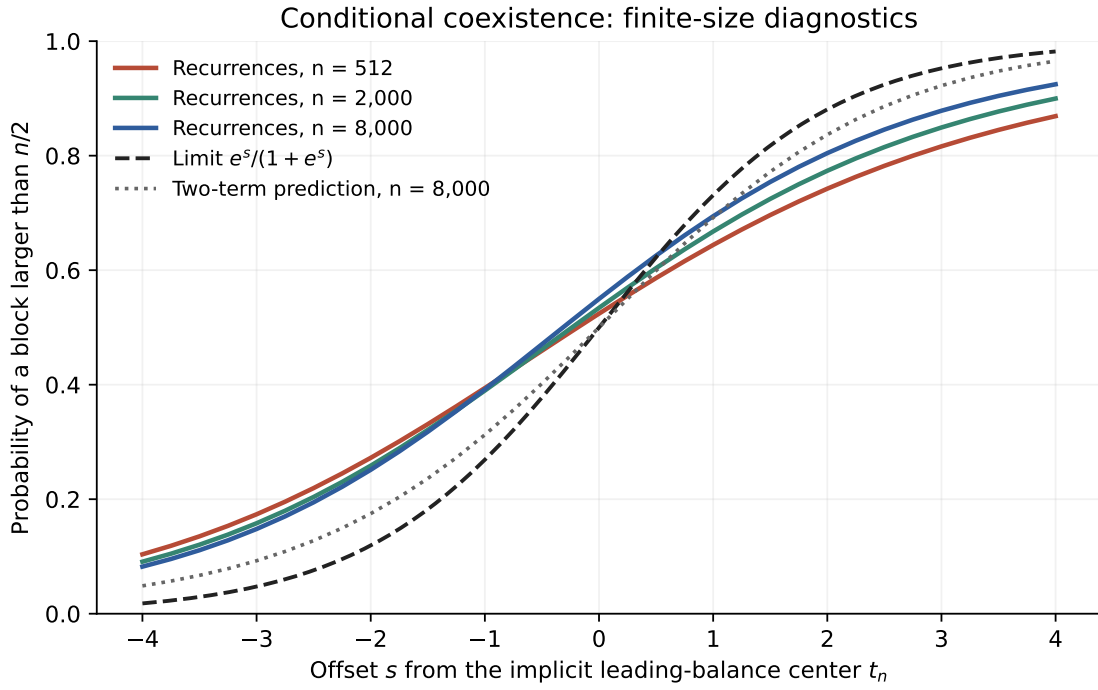


Figure 1: Slow finite-size behavior of the giant probability around the implicit center (120). The limiting logistic curve is shown for comparison. The finite curves are floating-point diagnostics, not interval certificates or evidence of monotone convergence.

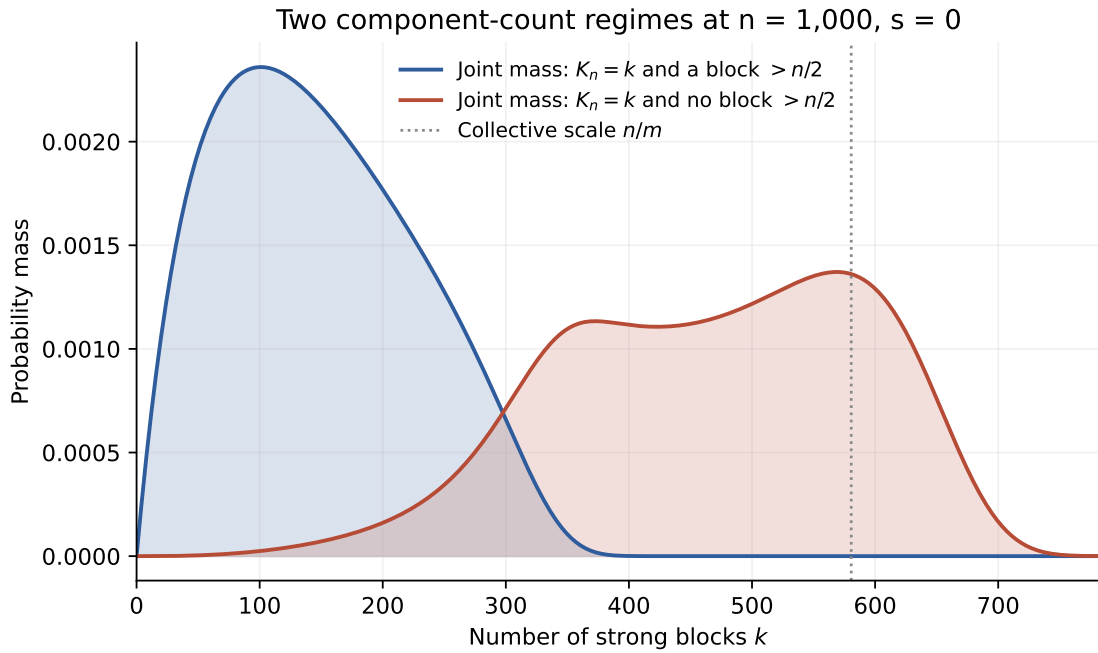


Figure 2: Block-count probabilities at $n = 1000$ and $\tau = \hat{b}_n$, separated by the giant event. The two curves are joint probabilities with their indicated events, so their combined mass is one. This finite distribution shows distinct count ranges and substantial corrections to the limiting scales.

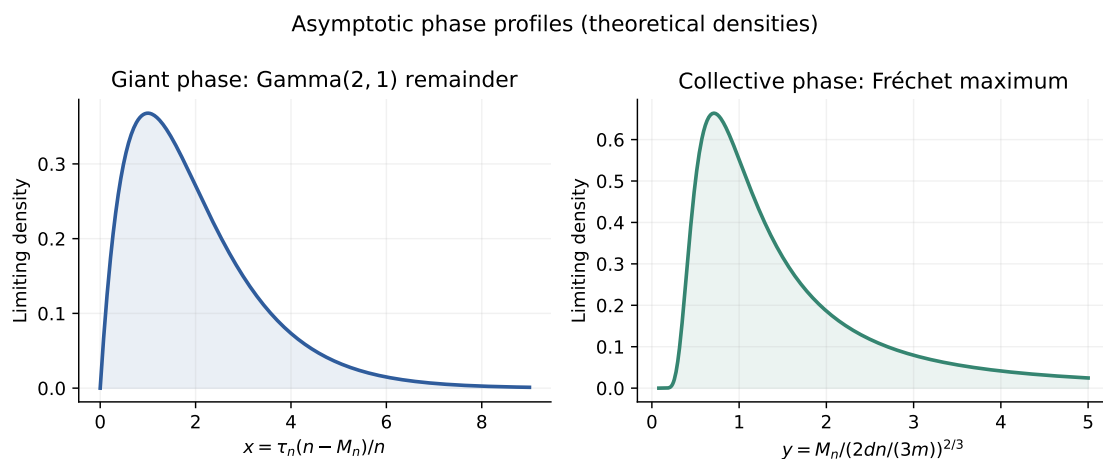


Figure 3: [Added in the write step, batch 85; caption from the delivered code README.] Theoretical limiting densities, not numerical distribution estimates: left, the Gamma(2, 1) density xe^{-x} of the rescaled giant deficit $x = \tau_n(n - M_{1,n})/n$ on $\mathcal{C}_n$ (Theorem 12.2); right, the Fréchet density $\frac{3}{2}y^{-5/2} \exp(-y^{-3/2})$ of the largest block on $\mathcal{F}_n$ in the variable $y = M_{1,n}/(2dn/(3m))^{2/3}$, which is V of (75) after the change of scale from a_n . The curves illustrate the proved limits; they are not an independent numerical test of them.

20 Further research questions

The following questions are not used as assumptions above. They continue the structural description at places where a new argument is actually needed.

- RQ1. A functional limit retaining the order of the blocks.** The joint count and ranked-size limits are now proved. Retain instead positions in the ordered sequence and study the centered cumulative block-size path on the whole unit interval, jointly with the count. Specify the topology, the endpoint conditioning, and the placement of large jumps. This would give a structural bridge to functional Gibbs-partition limit theorems.
- RQ2. A sharp finite-size coexistence center.** Continuity and uniform logistic convergence already imply that, for sufficiently large n , a weight u_n^* near the transition satisfies $\mathbb{P}_{n,u_n^*}(M_{1,n} > n/2) = 1/2$. Prove local uniqueness and derive sharper root corrections beyond (121). The implicit leading balance is not an exact median equation. Monotonicity of this particular event in the block weight should be proved rather than inferred from its limiting profile.
- RQ3. Rare configurations and unbounded residual observables.** Corollary 17.2 gives every fixed inverse-logarithmic order in total variation through an explicit count-polynomial tilt. Determine relative errors on specified rare residual configurations, or develop weighted-norm estimates controlling unbounded observables. These demands are stronger than total variation. For example, the unconditioned reference residual mass has an infinite second moment, whereas the actual residual mass is bounded by $n/2$. The polynomial tilt in the count alone does not remove this distinction.
- RQ4. Quantitative stable approximation.** Obtain explicit bounds for the conditional distribution in (74). The Fourier proof gives convergence but not an optimized rate. The exact next Puiseux coefficients may produce local corrections, but control outside fixed central windows must be retained.
- RQ5. Uniform growing shifts and the two pure regimes.** Extend the mixture theorem to $s = s_n \rightarrow \pm\infty$, identifying the largest ranges for which the phase-conditioned laws remain uniform. The compact- s subtraction in (112) is not uniform when one of the contributions becomes negligible. A proof must replace it with a relative estimate for the smaller event.
- RQ6. The critical and supercritical sides.** At $u = u_c$ the renewal mass is of order one, while for fixed $u > u_c$ a pole occurs inside the original convergence radius. Give a joint structural theorem connecting the logarithmic subcritical window, the critical law, and the moving supercritical pole. The present article does not analyze a negative τ .
- RQ7. General weights and slowly varying tails.** Prove the required local two-term coefficient estimate for broader classes $q_j = j^{-1-\alpha}L(j)$, and determine how L changes the center. Compare the exact hypotheses and conclusions with the Gibbs-partition and variable-box-number literature before asserting a new universality theorem.
- RQ8. Certified computation and exact sampling.** Use interval arithmetic for the constants and normalized recurrence to enclose finite-size phase probabilities. The product-string approximation also suggests a sampler on $\mathcal{C}_n$ with an explicit error budget; implement a rejection correction using $h_n(T)$ and prove its complexity.
- RQ9. Proof-assistant formalization.** Formalize the equality-index decomposition, the marked giant identity, and the total variation lemma first. They use finite combinatorics, geometric

series, and elementary expectation bounds. Then formalize the stable local limit and the common-contour estimate as separate analytic dependencies. Merely checking the integer table would not formalize any of the limit theorems.

A Notation and normalization checks

Symbol	Meaning
S_n, I_n	All score sequences and nonempty irreducible score sequences.
$S_{n,k}$	Number with exactly k irreducible blocks.
A, N_j	Logarithmic generating function and its divisor coefficients.
p	$I(1/4) = 1 - e^{-\lambda}$; not a phase probability.
$u, r = up$	Weight per irreducible block and its critical normalization.
q_j, Q	Probability mass function of a critical block length and its pgf.
m	Mean of q ; $m = \mu e^{-\lambda}/p$.
d	Tail amplitude: $q_j \sim dj^{-5/2}$.
κ	Puiseux and stable coefficient: $\kappa = d\Gamma(-3/2)$.
$Z_n(r)$	Normalized total weight $4^{-n}W_n(r/p)$.
$B_n(r)$	Unnormalized weight of the event $M_{1,n} > n/2$.
B_n	Second-maximum scale in (77), always without a parameter argument.
δ_n	$1 - r_n$, asymptotic to $m\tau_n/n$.
$b_n, \hat{b}_n$	Elementary and implicit leading-balance centers.
s, π_s	Bounded displacement from the center and limiting giant probability.
$K_n, M_{j,n}$	Number of blocks and ranked sizes.
a_n	$(\kappa n/m)^{2/3}$, used to normalize the fluid count.
G, Z	Gamma(2, 1) and the standard stable law (73).

Some checks prevent tempting normalization errors. First,

$$\Gamma(-3/2) = \frac{4\sqrt{\pi}}{3} > 0, \quad \frac{d}{m} = \frac{1}{2\mu\sqrt{\pi}}, \quad \frac{\kappa}{m} = \frac{2}{3\mu}.$$

Second, the block-count scale in the fluid theorem is a_n/m , not a_n ; inversion of a mean- m sum supplies the extra factor. Third, on $\mathcal{C}_n$,

$$\delta_n K_n \approx \frac{m\tau_n}{n} K_n, \quad \frac{\delta_n}{m} (n - M_{1,n}) \approx \frac{\tau_n}{n} (n - M_{1,n}).$$

Both converge to the same Gamma variable. Finally, the weights π_s and $1 - \pi_s$ are probabilities only after the exact large-block event has been identified.

B Dependency and validation record

The theorem dependencies are short enough to state without a large diagram. The exact identity (62) and the decomposition (60) yield the renewal model. The analytic split proves the local expansion and tail. The local expansion yields both the uniform two-term coefficient estimate and the Fourier local limit theorem. Independently, the tail and the finite mean yield the giant-deletion total

variation theorem. The total weight and the giant weight identify the mixture. Central local-limit windows exhaust its other component. The residual transforms then give all finer giant-phase laws. The exact enumerative theorem is the only substantial outside combinatorial input whose full proof is not reproduced. Standard complex integration, Fourier inversion, weak-convergence criteria, and the strong law of large numbers are used in their usual forms. No unproved probabilistic large-deviation estimate is inserted. The full source and the output records make the arithmetic and typesetting reproducible; analytic correctness rests on the proofs.

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